



MASTER'S THESIS

**Foundation-Model-Guided Zero-Shot
Synthesis of Human-Scene Interaction**

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Declaration

I declare that this thesis has been completed independently and that no sources or aids other than those listed have been used.

Munich, 20.07.2026

Place, Date

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Signature

Abstract

Generating a human who interacts convincingly with a real 3D scene is difficult because the input is usually semantic, while the output must satisfy geometry. A hand should meet the correct object surface, feet should support the body, and the body should not pass through nearby furniture or the room itself. This thesis studies a zero-shot pipeline for static human-scene interaction synthesis from a single RGB view, a ScanNet++ scene mesh, and a language instruction.

The method uses foundation models where their strengths are most useful, but keeps the final result tied to metric 3D geometry. First, a vision-language model converts the instruction into a Scene Interaction Graph that names the target object, relevant human body parts, and intended contact edges. An image editing model then inserts a plausible human into the scene, producing a 2D reference for the interaction. A second contact-inpainting step estimates object-surface contact masks for the body parts specified by the graph. These masks are projected from the image view onto the visible scene mesh and converted into 3D contact targets. Starting from an SMPL-X body initialized with GVHMR, the final stage optimizes pose, orientation, translation, and scale using contact, depth ordering, scene collision, self-intersection, height, and pose-prior terms.

The pipeline is evaluated on real indoor scenes using physical contact distances, non-collision metrics, render-based semantic consistency, and vision-language model verification. The result is a practical framework for turning text-level interaction intent into a geometrically grounded 3D human placement without task-specific training.

Keywords: human-scene interaction, foundation models, zero-shot synthesis, SMPL-X, contact optimization

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Thank advisors, supervisors, collaborators, funding sources, friends, and family here. Keep this section personal and direct.

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List of Abbreviations

HOI	Human-Object Interaction
ML	Machine Learning
RGB-D	Red, Green, Blue, and Depth

Chapter 1

Introduction

This chapter motivates zero-shot static 3D human-scene interaction synthesis from language, a calibrated RGB image, and a reconstructed scene mesh. It places the thesis in relation to learned HSI models, scene-constrained body optimization, and recent foundation-model-based systems, then states the problem formulation, approach, and contributions.

1.1 Why Human-Scene Interaction Matters

Indoor scenes are built around human action. Chairs are placed so that people can sit, tables so that people can work or eat, cabinets so that objects can be stored and retrieved, and handles so that drawers, doors, and appliances can be opened. The meaning of these objects is therefore not exhausted by their geometry or category labels. A chair is not only a shape occupying space; it affords support for a seated body. A drawer is not only a box-like structure inside a cabinet; it has a front face and a handle that invite a particular hand motion. A room becomes meaningful because human bodies can stand, reach, grasp, lean, sit, lie, and move through it.

This distinction is central to 3D scene understanding. Modern reconstruction methods can produce increasingly accurate models of indoor environments, including camera poses, dense geometry, object surfaces, and scene layout. However, a reconstructed scene by itself remains a passive description of the environment. It tells us where surfaces are, but not which surfaces can support the body. It shows object boundaries, but not which regions are functional for a task. It captures the state of a room, but not the ways in which a person could physically use that room. For applications that require human-centered reasoning, this missing layer is fundamental.

Human-scene interaction provides this missing layer by connecting scene structure to bodily action. Instead of asking only what objects are present, it asks how a person can engage with them. Where should the feet be placed when reaching for a cabinet? Which part of

a sofa should support the pelvis and back during sitting? Where should the hand make contact when opening a drawer? What body orientation is plausible when lifting a trash bin? These questions require reasoning jointly about object function, human body parts, contact, balance, reachability, and spatial constraints. They cannot be answered by object recognition or scene reconstruction alone.

This capability is important for a broad class of downstream systems. In embodied AI and robotics, an agent must reason not only about object locations, but also about how humans act around those objects and which parts of the scene are functionally relevant. In augmented and virtual reality, inserted humans must appear grounded in the environment rather than floating near objects or intersecting with them. In animation, synthetic data generation, and digital twins, realistic human activity requires bodies that are placed according to the affordances of the scene. In human-centered scene analysis, the presence or absence of plausible interactions can reveal whether a reconstructed environment is usable, accessible, or semantically coherent.

The challenge is therefore not simply to add a human body to a 3D scene. A visually plausible body can still be physically or functionally wrong: a seated person may hover above a chair, a hand may reach the side of a cabinet instead of the drawer handle, or a body may intersect with furniture while appearing reasonable from one camera view. Human-scene interaction requires a stronger form of consistency. The body must be semantically aligned with the requested action, geometrically aligned with the scene, and physically plausible with respect to contact and collision.

This thesis studies human-scene interaction synthesis as a step toward scene representations that are not only geometric, but actionable. The goal is to synthesize human bodies that express how a real indoor scene can be used. By grounding human pose in scene affordances, object function, and physical contact, HSI synthesis moves beyond static reconstruction toward a more complete understanding of indoor environments as spaces for human activity.

1.2 From Reconstructed Geometry to Interaction Synthesis

A 3D reconstruction converts visual observations of a room into a metric representation of space. It provides camera poses, scene geometry, object surfaces, and the spatial layout of the environment. This representation is essential for grounding perception in real-world coordinates: it tells us where the floor is, how far a chair is from a table, where a cabinet is located, and which surfaces are visible or occluded. However, reconstruction alone does not specify how a human body should be arranged in the scene. Geometry describes the shape of the environment, but interaction requires reasoning about the relationship between that geometry and the body.

This gap becomes apparent even for simple activities. To synthesize a person sitting on a chair, it is not enough to know that a chair exists. The body must be placed so that the pelvis and thighs are supported by the seat, the feet remain near the floor, and the torso has a plausible orientation relative to the chair. To synthesize a person opening a drawer, it is not enough to locate the cabinet. The hand should reach the drawer pull or handle, while the body should stand at a reachable distance and face the object in a plausible way. To synthesize a person lifting a trash bin, the hands, arms, torso, and feet must form a coherent configuration around the object. In each case, the interaction is defined not only by the object category, but by the spatial and functional relation between body parts and scene parts.

Human-scene interaction synthesis addresses this problem by generating a human body configuration that is jointly consistent with an action and a 3D scene. Unlike image-based human insertion, the goal is not only to make a rendered view appear plausible. The synthesized body must exist in the same coordinate frame as the scene and must satisfy constraints that are meaningful in 3D: contact should occur at appropriate locations, the body should have a plausible scale and orientation, and severe interpenetration with the environment should be avoided. The output is therefore a scene-grounded human state, not simply a visual overlay.

This distinction separates interaction synthesis from several related tasks. Object detection identifies what is present in the scene, but it does not determine how the object can be used. Human pose estimation recovers a body configuration from observations, but it usually assumes that the person is already present. Text-to-image generation can produce visually convincing depictions of people interacting with objects, but the result is not necessarily metrically consistent with a reconstructed scene. Human-scene interaction synthesis combines aspects of all three problems while adding a stronger requirement: the generated body must be semantically aligned with the requested action and physically grounded in the 3D environment.

The difficulty is that interaction is structured. A single instruction often implies a set of relations among the action, the object, the relevant object part, the participating body parts, and the supporting scene surfaces. Sitting involves support relations between the lower body and a seat. Opening involves a reaching or grasping relation between a hand and a functional element. Lifting involves coordinated contacts between the hands and the object, together with a stable stance on the floor. These relations are not directly given by the reconstruction. They must be inferred from the task and then realized as a plausible body placement in the scene.

Interaction synthesis can therefore be viewed as the step from a passive scene model to an actionable scene model. A passive model answers geometric questions such as where surfaces are and how objects are arranged. An actionable model supports questions about what a human can do in the environment, where the body should be placed, which body parts

should contact which scene regions, and whether the resulting configuration is physically credible. This thesis focuses on this transition: synthesizing a human body that does not merely appear inside a reconstructed room, but uses the room in a way that is consistent with language, geometry, and physical contact.

1.3 The Zero-Shot Challenge

Human-scene interaction synthesis becomes especially difficult when the system is expected to operate in a zero-shot setting. In a closed setting, the model may assume a fixed set of object categories, a fixed set of actions, or training examples that closely resemble the test scenes. Under these assumptions, the problem can be approached as learning regularities from paired examples of humans interacting with objects. The model can learn, for instance, that people sit on chairs, stand on floors, touch tables with their hands, or lie on beds. However, such regularities are only reliable when the test interaction lies within the distribution of the training data.

Real indoor scenes do not satisfy this assumption. They contain diverse object instances, cluttered layouts, partial visibility, unusual viewpoints, and functional elements that vary widely in appearance. A drawer handle may be horizontal, vertical, recessed, metallic, wooden, large, small, or partially occluded. A chair may be an office chair, dining chair, stool, sofa, or bench. Even when the object category is familiar, the region relevant to an interaction may differ across instances. The same action phrase can therefore require different geometric solutions depending on the scene. ‘Open the drawer’ requires the hand to reach the handle of a particular drawer in a particular cabinet. ‘Sit on the chair’ requires the body to find a support surface of the correct height and orientation. ‘Lift the bin’ requires a graspable region on the visible object and a stable full-body configuration around it.

This variability exposes the limitation of relying only on supervised 3D human-scene interaction data. Paired HSI data is expensive to collect because it requires reconstructed scenes, accurate human body fits, object annotations, action labels, contact information, and enough diversity in body shapes, object instances, and room layouts. As a result, existing datasets cover only a small subset of the possible interactions that occur in indoor environments. Learned HSI models can produce strong results for common interactions represented in their training sets, but they often depend on closed vocabularies, predefined object categories, known target objects, or interaction patterns observed during training. These constraints make it difficult to synthesize interactions for arbitrary language instructions in previously unseen reconstructed scenes.

Recent work has therefore explored the use of large vision-language, image-generation, and video-generation models as sources of interaction knowledge. These models have seen

broad visual and linguistic evidence of people using objects, and they can often suggest plausible body poses, object affordances, and interaction layouts without being trained directly on 3D HSI datasets. This has motivated zero-shot and training-free approaches to human-scene interaction synthesis, where foundation models provide semantic or visual priors and 3D geometry provides metric grounding. Methods such as GenZI, FunHSI, and ZeroHSI show that this direction is promising because interaction knowledge can be transferred from large-scale visual models to 3D scenes without requiring paired 3D interaction supervision.

However, zero-shot HSI is not solved by generating a plausible image of a person interacting with an object. A generated image may express the intended action visually, but it does not guarantee that the body is correctly placed in the reconstructed scene. The hands may appear near an object in image space while missing the corresponding 3D surface. The body may look plausible from one camera view while penetrating furniture or floating above the floor. The generated interaction may also contact the wrong part of the object: the side of a cabinet instead of the drawer handle, the rim of a bin instead of a graspable side, or the back of a chair instead of the seat. The challenge is therefore to convert broad interaction priors into scene-specific, metrically valid human-body configurations.

The zero-shot setting thus requires solving several coupled questions. Given an open-vocabulary instruction, the system must infer which scene object is relevant, which part of that object affords the action, which human body parts should participate, where supporting contacts should occur, and how the full body should be arranged in 3D. These questions cannot be answered independently. The choice of contact region affects the pose; the pose affects the feasible body location; the body location affects reachability, balance, and collision; and the scene geometry constrains all of these decisions. Zero-shot HSI is therefore not simply a problem of text-conditioned pose generation, but one of grounding an inferred interaction structure in a specific reconstructed environment.

This thesis addresses the zero-shot challenge in the static 3D setting. Rather than assuming a fixed action vocabulary or paired 3D interaction examples for every task, it treats the instruction as an open-ended description of how a person should use the scene. The goal is to infer the relevant interaction structure and synthesize a single SMPL-X body that realizes that structure in the original scene coordinate frame. This formulation keeps the problem narrower than full dynamic interaction generation, but it preserves the central difficulty of zero-shot HSI: the system must transform language and scene observations into a physically plausible human-scene contact configuration without relying on task-specific 3D supervision.

1.4 Body-Part–Scene-Part Grounding

The previous sections framed human-scene interaction synthesis as more than placing a plausible body near a relevant object. The remaining question is what structure should connect the language instruction, the human body, and the scene. A natural first answer is to reason at the object level: a person sits on a chair, opens a drawer, touches a table, or lifts a bin. Object-level reasoning is useful, but it is not precise enough for interaction synthesis. Most interactions are not defined by whole objects. They are defined by relations between specific parts of the human body and specific regions of the scene.

Consider the instruction ‘a person opening a drawer.’ The target object may be a cabinet or drawer unit, but the interaction itself is concentrated at the drawer handle or front edge. The relevant human part is usually a hand, not the entire body. At the same time, the rest of the body is not arbitrary: the feet must remain supported by the floor, the torso must be oriented toward the cabinet, and the arm must be arranged so that the hand can reach the functional part. Similarly, ‘a person sitting on a chair’ primarily requires support between the lower body and the seat, but also implies a plausible relation between the feet and the floor and between the torso and the chair orientation. These examples show that an interaction is best understood as a set of body-part–scene-part relations.

This thesis adopts that view. Instead of treating the human body as a single object and the scene as a collection of object categories, it represents an interaction through participating body regions, scene regions, and contact relations between them. The human body is decomposed into interaction-relevant parts, such as hands, arms, feet, pelvis, thighs, back, and torso. These parts correspond to regions on the SMPL-X surface that commonly participate in physical interactions. On the scene side, the relevant regions may be broad support surfaces, such as floors, seats, beds, and tabletops, or smaller functional elements, such as handles, knobs, switches, faucets, and drawer pulls. The task of HSI synthesis is then to arrange the body so that the appropriate body regions are grounded to the appropriate scene regions.

This formulation is important because different actions require different kinds of contact. Support interactions, such as sitting, lying, standing, or leaning, depend on stable contact between load-bearing body regions and supporting scene surfaces. Manipulation interactions, such as opening, pulling, pressing, or lifting, depend on localized contact between the hands and functional object parts. Many interactions combine both types. A person opening a drawer must stand on the floor while reaching the handle. A person lifting a bin must grasp the object while maintaining a stable stance. A person leaning over a sink may contact the floor with the feet, the counter with the hands or torso, and the faucet with one hand. Representing these interactions as structured body-scene contacts makes these dependencies explicit.

The body-side representation also matters. SMPL-X provides a detailed parametric human

mesh, but the mesh vertices themselves are not semantically meaningful unless they are grouped into interaction regions. This thesis therefore uses manually defined contact regions on the SMPL-X body surface corresponding to interaction parts such as hands, arms, feet, pelvis, back, and other task-relevant regions. These regions provide a bridge between high-level language and low-level geometry: an instruction may mention a hand, a seated posture, a leaning action, or a grasping motion, while the final synthesis must place specific surface regions of the body near specific surface regions of the scene.

The scene-side representation is equally important. For many tasks, detecting the object category or even the object part is insufficient. The same cabinet can support several different interactions: opening a drawer, reaching for an upper shelf, leaning on the counter, or standing beside it. Each task selects a different scene region and a different set of body regions. Conversely, the same body part can participate in different functional roles: a hand can grasp a handle, press a switch, support the body on a tabletop, or hold the side of a bin. Interaction synthesis must therefore resolve not only which object is involved, but which object region is relevant and what type of body contact it requires.

Moreover, part-level grounding alone does not fully determine the interaction. Many object parts are spatially extended, and the exact location of contact within a detected part can strongly affect the plausibility of the synthesized pose. For example, when lifting a table, it is not enough to identify the tabletop as the relevant part; the hands should contact plausible grasp or support locations on the tabletop, often near accessible edges or corners rather than at arbitrary points on the surface. Similarly, for an interaction involving a ladder, the system must reason not only about the ladder as an object, or about its rails and rungs as parts, but also about which specific rail and rung regions should be contacted by the hands and feet. The same rung may support a foot at one location while a hand grasps a nearby rail at another, and these local choices determine whether the resulting pose is physically and functionally coherent.

This distinction is important because semantic part labels provide structure, but they do not by themselves specify contact geometry. A tabletop, rail, rung, seat, counter, cabinet front, or floor region can contain many possible contact locations, only some of which are compatible with the requested action, the human pose, the object affordance, and the surrounding scene. Body-part-scene-part grounding therefore requires a finer notion of localization: the framework must identify not only the relevant object and object part, but also the appropriate contact region within that part. This finer grounding allows the synthesized body to interact with the scene in a way that is both semantically correct and geometrically plausible.

To capture this structure, the thesis uses a Scene Interaction Graph as an intermediate representation. The graph expresses the target scene element, the participating human interaction parts, and the expected contact or support relations between them. Its purpose is not to describe every detail of the final pose, but to specify the interaction constraints

that the pose should satisfy. In this sense, the graph acts as a compact bridge between open-language task descriptions and 3D body synthesis. It turns an instruction such as “lift the trash bin with both hands” into a structured interaction hypothesis involving the bin, the two hands, the arms, the feet, and the supporting floor.

A key aspect of this thesis is that the scene-side contact regions are not assumed to be given by manual annotations, object geometry alone, or part labels alone. Instead, they are inferred through image-based generative reasoning. Starting from the scene image, a diffusion-based inpainting model first synthesizes a plausible human performing the requested interaction in the scene. This generated human provides a visual hypothesis about pose, orientation, laterality, and approximate contact layout. Then, conditioned on the interaction parts specified by the Scene Interaction Graph, a second diffusion-based inpainting step marks the corresponding contact regions on the target object or support surface in the original scene image. In this way, the scene image is used twice: first to imagine how the interaction should look, and then to localize where the relevant body parts should make contact on the object or scene.

This image-based contact grounding is particularly valuable for functional interactions, where the relevant contact region may be small, localized, or ambiguous even when the correct object part is known. A cabinet may contain many possible surfaces, but only the handle or drawer pull is appropriate for opening. A trash bin may have several visible faces, but only some are plausible grasp regions for lifting. Likewise, a large tabletop, ladder rail, or ladder rung may be semantically correct at the part level, but the interaction still depends on where within that part the hands, feet, or other body regions should make contact. By using diffusion models to propose both the human interaction hypothesis and the object-side contact regions, the framework leverages broad visual priors while keeping the final interaction tied to the actual scene and to the fine-grained spatial layout of the interaction.

Body-part–scene-part grounding is therefore the central abstraction behind this thesis. It makes the interaction explicit enough to guide synthesis, but flexible enough to support open-vocabulary instructions and previously unseen scenes. By grounding actions in relations between human surface regions and scene surface regions, and by further localizing contact within those scene regions, the formulation moves beyond object-level and part-level semantics toward a representation of how the body physically and functionally engages with the environment.

1.5 Problem Setting and Contributions

This thesis studies zero-shot static 3D human-scene interaction synthesis in real reconstructed indoor scenes. The input consists of a natural-language instruction, a calibrated

RGB image of the scene, and a reconstructed scene mesh. The output is a single SMPL-X body placed in the original scene coordinate frame. The synthesized body should express the requested action, contact the appropriate scene regions with the appropriate body parts, and remain physically plausible with respect to scale, orientation, support, and collision.

The setting is zero-shot because the method is not trained on paired 3D examples of the requested interaction in the target scene. The instruction may describe common support interactions, such as sitting on a chair, lying on a bed, or leaning on a table, or functional interactions, such as opening a drawer, pressing a switch, turning a faucet, or lifting a bin. In each case, the system must infer the relevant scene element, the functional or supporting region on that element, the participating human body parts, and the spatial arrangement required to realize the interaction. The target object and the required contact regions are not assumed to be available as ground-truth annotations.

The formulation is static. The goal is not to synthesize a full motion sequence, simulate forces, or modify the state of the scene. Instead, the output represents a single interaction state. For steady actions such as sitting or lying, this state corresponds to the interaction itself. For actions involving movable or articulated objects, such as opening a drawer, pulling a door, or lifting a trash bin, the synthesized body represents the pre-action or contact-ready state: the human has reached the relevant scene region and adopted a plausible pose, but the object has not yet moved. This choice isolates a fundamental subproblem of human-scene interaction: determining where contact should occur and how the body should be configured in a real 3D environment.

The proposed approach is built around body-part-scene-part grounding. An open-language instruction is first interpreted as a structured interaction involving human interaction regions, scene regions, and contact relations. On the human side, SMPL-X is treated not only as a parametric body model, but as a surface with semantically meaningful interaction regions, such as hands, arms, feet, pelvis, thighs, back, and torso. On the scene side, the relevant regions may be broad support surfaces or localized functional elements. The synthesis problem is then to place the body so that these body regions and scene regions are brought into physically and semantically meaningful correspondence.

To support this grounding in open-vocabulary scenes, the thesis combines structured interaction reasoning with generative visual priors and scene-aware 3D fitting. A Scene Interaction Graph represents the expected relations among the action, target scene element, participating body parts, and contact regions. Diffusion-based image reasoning provides visual hypotheses about how the interaction should appear and where the corresponding object-side contacts should lie. These hypotheses are then grounded in the reconstructed scene to synthesize a metric SMPL-X body that satisfies the inferred interaction structure. At a high level, the method uses foundation models to reason about what interaction is plausible, while using the scene reconstruction to ensure that the final body is placed in

3D rather than merely depicted in an image.

This thesis makes the following contributions:

1. **A zero-shot formulation of static 3D human-scene interaction synthesis.** The thesis defines the task of synthesizing a single SMPL-X body in a real reconstructed indoor scene from an open-language instruction, a calibrated RGB image, and a scene mesh, without requiring paired 3D interaction examples for the target task.
2. **A body-part–scene-part grounding view of HSI.** The thesis formulates interaction synthesis as the problem of grounding relations between human interaction regions and scene interaction regions, rather than only matching whole-body poses to object categories.
3. **A Scene Interaction Graph representation for open-vocabulary interactions.** The thesis introduces a structured representation that captures the target scene element, participating human body parts, scene regions, and expected contact or support relations implied by a natural-language instruction.
4. **A SMPL-X interaction-part abstraction for contact-aware synthesis.** The thesis defines semantically meaningful contact regions on the SMPL-X body surface, enabling high-level interaction descriptions involving hands, arms, feet, pelvis, back, and other body regions to be connected to low-level 3D body geometry.
5. **An image-guided strategy for localizing scene-side contact regions.** The thesis uses generative image reasoning to infer where the human interaction parts should contact the target object or support surface, allowing functional regions such as handles, knobs, drawer pulls, and graspable object areas to be identified without manual scene-side contact annotation.
6. **A scene-grounded synthesis framework for physically plausible body placement.** The thesis synthesizes a full SMPL-X body in the reconstructed scene coordinate frame, using the inferred interaction structure and scene geometry to encourage semantic consistency, appropriate contact, realistic scale, and reduced physical artifacts such as floating or penetration.

Together, these contributions move static HSI synthesis beyond object-level placement and toward interaction-level grounding. The thesis argues that plausible human-scene interaction requires knowing not only what object is involved, but which parts of the human body should engage which parts of the scene, and how that engagement can be realized as a coherent 3D body configuration.

1.6 Thesis Structure

Chapter 2 reviews the foundations and related work necessary for this thesis, including 3D scene reconstruction, parametric human body models, human-scene interaction synthesis, affordance and contact reasoning, foundation models for visual generation, and evaluation of physically grounded human-scene configurations.

Chapter 3 presents the proposed zero-shot static HSI synthesis framework. It describes the Scene Interaction Graph, the SMPL-X interaction-part abstraction, diffusion-based human and contact-region reasoning, scene-side contact grounding, and the final scene-aware SMPL-X fitting procedure.

Chapter 4 defines the experimental setup, including the ScanNet++ scenes, interaction instructions, selected action categories, implementation details, baselines or ablations, and quantitative and qualitative evaluation metrics.

Chapter 5 reports the experimental results and analyzes the behavior of the method. It examines semantic consistency, contact accuracy, physical plausibility, pose diversity, qualitative examples, ablation studies, and common failure modes.

Chapter 6 summarizes the findings of the thesis and discusses limitations and future directions, including dynamic interaction synthesis, richer physical reasoning, multi-view grounding, multi-human interactions, and more complete modeling of object state changes.

Chapter 2

Related Work

This chapter positions the thesis within prior work on human-scene interaction synthesis. It organizes the literature along two axes that are central to the thesis: trained versus zero-shot methods, and static 3D versus dynamic 4D interaction synthesis.

2.1 Contact Representations and Supervision for 3D HSI

Human-scene interaction requires more than placing a plausible human pose inside a 3D environment. A synthesized body must also be geometrically compatible with the surrounding scene: it should be supported by appropriate surfaces, avoid severe interpenetration, and make contact with scene regions that are meaningful for the intended interaction. This distinction is especially important for static 3D HSI, where the output is a single body configuration rather than a motion sequence. In such a setting, contact provides one of the main cues that separates a merely plausible pose from a physically and functionally grounded interaction.

The scene representation determines whether such contact can be enforced metrically. Indoor reconstruction datasets such as ScanNet and ScanNet++ provide RGB observations, camera calibration, and reconstructed scene geometry, allowing image-space evidence to be related to a common 3D coordinate frame [1, 2]. This calibration is essential for methods that reason jointly in image and scene space. A contact cue observed in an RGB view is only useful for 3D synthesis if it can be associated with a surface in the reconstructed scene. In this thesis, ScanNet++ provides the calibrated image, camera parameters, and scene mesh on which body-scene contact constraints are ultimately enforced.

On the human side, parametric body models provide a differentiable representation for optimization. SMPL-X is particularly suitable for human-scene interaction because it represents the full body as a posed mesh with articulated body, hand, face, shape, and global transformation parameters [3]. Its dense surface makes it possible to define contact

not only at sparse joints, but over anatomically meaningful body regions. For example, grasping, standing, leaning, sitting, and pushing involve different parts of the body surface. Therefore, contact-aware HSI methods often require a mapping from semantic body parts to subsets of body vertices, rather than treating the body as an undifferentiated mesh.

Scene-aware human fitting demonstrated early that image-based human estimates are often physically invalid when evaluated in 3D. PROX extended monocular SMPL-X fitting with constraints from reconstructed scenes, using contact and interpenetration terms to reduce ambiguities in human pose estimation [4]. Its central insight is directly relevant to synthesis: a body may explain the image evidence while still floating above the floor, penetrating furniture, or failing to touch a supporting surface. By introducing the scene as an active constraint, PROX established contact and collision avoidance as fundamental components of 3D human-scene reasoning.

Subsequent work developed contact into a learned representation for HSI. PLACE models proximity and contact between an articulated human body and a 3D scene, emphasizing that realistic bodies in scenes depend on local body-scene relationships rather than only global placement [5]. POSA further represents interaction in a body-centric way by augmenting SMPL-X vertices with contact probabilities and semantic scene labels [6]. These methods show that contact can serve as an intermediate representation between human pose and scene affordance. They also motivate body-surface reasoning: plausible HSI depends not only on whether a person is near an object, but on which body surface is near or touching which scene surface.

Dense contact datasets and predictors further highlight the importance of body-level contact supervision. RICH introduced high-quality human-scene capture with dense vertex-level body-scene contact labels, and BSTRO uses this data to infer body-scene contact from a single image [7]. DECO extends dense 3D contact estimation to in-the-wild images by predicting vertex-level contact on the human body using both body-part and scene-context cues [8]. These works are important because they move beyond coarse contact categories and sparse joints. They treat contact as a dense surface property, which is more appropriate for physical interaction than global action labels alone.

Human-object interaction datasets provide another source of contact supervision. GRAB captures whole-body grasping motions with detailed body, hand, object, and contact information [9]. BEHAVE provides multi-view RGB-D recordings with 3D human and object fits, together with contact annotations for full-body human-object interaction [10]. These datasets are valuable because they capture how the full body participates in object interaction, not only the hands. However, they are primarily object-centric. The target object is usually explicitly present and tracked, and the interaction is observed or captured rather than inferred from an open-vocabulary instruction in an arbitrary reconstructed room.

Together, these works establish that contact is a central representation for 3D HSI. They also reveal several limitations that matter for this thesis. First, many methods model contact primarily on the human side, for example by predicting which SMPL or SMPL-X vertices are in contact. This is useful for understanding the body, but it does not by itself identify the exact scene surface that should be contacted in a new reconstructed environment. Second, contact supervision is often tied to specific datasets, object categories, scenes, or captured interaction types. This limits generalization to open-vocabulary instructions and unseen scene layouts. Third, object-level grounding is often too coarse. Even when the correct object is known, the method must still determine the appropriate contact patch: a hand lifting a table should contact a plausible region of the tabletop or edge, a hand opening a washing machine should contact the door or handle region, and a foot climbing a ladder should contact a rung or step rather than the object instance as a whole.

This thesis builds on the contact-centered view of HSI, but uses contact differently. Instead of learning a contact prior from paired human-scene interaction data, the method defines body-side contact regions on SMPL-X and estimates the corresponding scene-side contact regions from visual interaction evidence. The scene-side target may be a functional part, such as a handle or control, but it may also be a pose-dependent contact patch on a larger surface, such as a tabletop, sofa seat, wall, rail, or floor. The resulting image-space masks are projected into the reconstructed scene mesh and used as explicit 3D contact targets for optimization. In this way, the method inherits the geometric accountability of contact-aware HSI while addressing a broader problem: open-vocabulary, body-part-specific scene-side contact grounding in calibrated reconstructed scenes.

2.2 Static and Language-Controlled HSI Synthesis

While contact representations define how human-scene compatibility can be described, HSI synthesis methods address the generative problem: producing a human body or motion that is plausible in a given scene. In the static 3D setting, the goal is to synthesize a single body configuration that is semantically meaningful and physically feasible with respect to the environment. This requires deciding where the body should be placed, how it should be oriented, which pose it should take, and how it should relate to nearby scene surfaces. Compared with scene-agnostic pose generation, static HSI synthesis must account for both the semantic role of the scene and its metric geometry.

Early work on 3D human affordance prediction framed this problem as estimating which human poses are supported by an indoor environment. Putting Humans in a Scene predicts human affordances in 3D indoor scenes by learning what poses are plausible for a given scene, such as sitting on a chair or standing on the floor [11]. The method combines semantic knowledge from 2D human poses with geometric knowledge from 3D indoor scenes, allowing it to predict semantically plausible and physically feasible human configurations. This

line of work is important because it treats the scene as an active source of affordances rather than as a passive background. However, the interaction is still controlled indirectly through learned affordance distributions, not through open-ended language instructions or body-part-specific contact requirements.

Generating 3D People in Scenes without People further develops static HSI synthesis by generating full SMPL-X bodies in empty 3D scenes [12]. The method first predicts semantically plausible body configurations conditioned on a latent scene representation and then refines the generated bodies with scene constraints to improve physical feasibility. This two-stage structure is influential: a learned model proposes a plausible human-scene configuration, while a refinement stage enforces physical constraints such as contact support and non-penetration. The method demonstrates that realistic static HSI requires both semantic compatibility and geometric correction. Nevertheless, the generated interactions are primarily governed by learned scene affordances rather than by a user-specified, open-vocabulary interaction with explicit contact intent.

Contact-aware synthesis methods such as PLACE and POSA also contribute to static HSI generation. PLACE uses local scene geometry to synthesize articulated humans that are compatible with nearby surfaces, emphasizing proximity and contact between the body and the environment [5]. POSA learns a body-centric representation of contact and scene semantics on SMPL-X, allowing plausible body placements to be generated or refined according to predicted body-scene relationships [6]. These methods show that contact is not only useful for evaluating a generated body, but can also guide synthesis itself. However, because the contact priors are learned from available interaction data, their behavior is tied to the distribution of scenes, poses, and contact configurations seen during training.

Semantic control is introduced more explicitly in COINS, which synthesizes humans interacting with 3D scenes from action-object specifications [13]. Instead of generating a plausible body anywhere in the scene, COINS allows the user to specify interactions such as sitting on a chair or touching a table. This is an important step from generic scene population toward controllable HSI synthesis. COINS also supports compositional interaction specifications, making it possible to combine simpler action-object relations into more complex human-scene configurations. However, the control signal remains a predefined action-object pair. This is suitable for interactions whose geometry is largely determined by the object category and action label, but it is less expressive for instructions that require fine-grained contact reasoning, such as selecting where on a large table surface the hands should touch, or which visible part of an object should be contacted.

The distinction between object-level control and contact-region control is important for this thesis. A semantic specification such as ‘touch table’ or ‘sit chair’ identifies the target object and broad action, but it does not necessarily determine the exact scene-side contact patch. Even when the relevant object part is large, such as a tabletop, sofa seat, wall, or floor, the precise contact region should still be consistent with the intended pose, body

scale, orientation, and scene layout. For functional interactions, the required contact may be even more localized, such as a handle, rail, rung, button, or edge. Therefore, static HSI synthesis requires more than choosing a plausible object and body pose; it may also require grounding each relevant body part to a specific region of the scene surface.

Language-conditioned HSI extends semantic control beyond fixed action-object labels, but most existing work in this direction focuses on motion rather than a single static body configuration. HUMANISE introduced language-conditioned human motion generation in 3D scenes by aligning motion sequences with indoor scenes and language descriptions [14]. This formulation shows that natural language can specify not only the action but also the interacting object and spatial intent. TextSceneMotion further decomposes scene-aware text-to-motion generation into target-object grounding and object-centric motion generation [15]. This decomposition is relevant because it separates the problem of understanding the language target from the problem of generating the human behavior around that target. TeSMo similarly studies text-controlled scene-aware motion generation using diffusion models, producing motions such as navigation and object interaction in 3D scenes [16].

Other dynamic HSI methods broaden the scope from static placement to long-horizon scene-aware behavior. SceneDiffuser formulates scene-conditioned generation, optimization, and planning as a diffusion process applicable to tasks such as human pose generation, human motion generation, grasping, and navigation [17]. TRUMANS scales dynamic HSI supervision by capturing diverse human interactions in indoor scenes and training a model for scene-conditioned interaction sequences [18]. These works are important because they show how language, motion priors, scene context, and contact can be integrated for temporally evolving interactions. However, their central problem is different from the one considered in this thesis. They must model trajectories, transitions, temporal coherence, and changing contacts over time, whereas the present work focuses on one static interaction state.

Overall, static and language-controlled HSI synthesis has progressed from affordance-based human placement, to contact-aware body generation, to semantic and language-conditioned interaction control. These methods establish that plausible HSI depends on both scene semantics and physical compatibility. However, most data-driven approaches remain constrained by learned interaction distributions, predefined action-object labels, or paired scene-motion supervision. They also typically ground interaction at the level of an object, action, or motion trajectory, rather than explicitly estimating the body-part-specific scene regions that should participate in contact. This leaves two coupled challenges for zero-shot static 3D HSI: first, extracting interaction-specific 2D contact regions on the scene from visual and language evidence; and second, converting those regions into metric 3D contact targets that can guide body optimization. The next section discusses foundation-model-based methods that address parts of this problem, and motivates the need for an explicit contact-localization and geometric-grounding stage.

2.3 Foundation-Model and Zero-Shot Interaction Generation

The methods discussed so far rely primarily on learned interaction priors, paired human-scene data, predefined action-object labels, or motion supervision. Recent foundation-model-based approaches follow a different strategy: instead of training a task-specific HSI model from paired 3D interaction data, they use pretrained language, vision-language, image generation, video generation, segmentation, and human reconstruction models as sources of interaction knowledge. These models have absorbed broad visual and semantic regularities about people, objects, actions, and everyday scenes. As a result, they can provide useful priors for interactions that are difficult to cover exhaustively with captured 3D HSI datasets.

In zero-shot HSI, foundation models are typically not used as final geometric solvers. A language model may infer the likely interaction structure from a text prompt, a vision-language model may judge whether a generated image satisfies an instruction, and an image generation model may synthesize a plausible human interacting with the scene. However, these outputs are usually expressed in language or image space. They do not directly provide a body mesh in the scene coordinate frame, nor do they automatically identify which reconstructed scene vertices should be contacted by which body parts. Therefore, zero-shot HSI requires an additional grounding step that converts foundation-model hypotheses into metric constraints.

GenZI is a key example of zero-shot static 3D HSI [19]. Given a 3D scene, an interaction description, and a specified interaction location, GenZI uses large vision-language and image inpainting models to generate plausible 2D human-scene interaction hypotheses from rendered scene views. These generated images provide visual evidence about the human pose and the intended interaction, after which a 3D human body is optimized in the scene. GenZI demonstrates that pretrained 2D models contain useful priors for 3D human-scene interaction, even without training on paired 3D HSI data. At the same time, the generated image mainly acts as an implicit interaction hypothesis. The method does not explicitly extract body-part-specific scene-side contact masks and use them as direct metric targets for optimization.

FunHSI is more directly concerned with open-vocabulary functional interaction [20]. It observes that many interaction prompts cannot be solved by object category alone: a person increasing room temperature, opening a window, or using a particular appliance must reason about the functional elements of the scene and the body configuration required to interact with them. FunHSI uses foundation models to identify functional scene elements, construct contact graphs, synthesize a human performing the task, and refine the 3D body through optimization for physical and functional correctness. This is closely related to the motivation of this thesis because both approaches treat open-vocabulary interaction as a contact-grounding problem rather than merely as object-level placement.

The present thesis differs in the way the scene-side contact is grounded. Instead of relying primarily on explicit functional-element identification, the proposed method first represents the interaction as body-part-to-scene contact requirements through a Scene Interaction Graph. The scene node may be a target object or the floor, and the precise contact patch is inferred later from visual evidence. This formulation covers functional contacts, such as touching a handle or control, but it also covers broader pose-dependent contacts, such as choosing where on a tabletop, wall, sofa seat, treadmill rail, or floor the body should touch. Thus, the problem is not only functional object-part discovery. It is broader: interaction-specific scene-side contact localization for each required body part.

This distinction is important because the difficult step is not only projecting 2D contact evidence into 3D. Projection is necessary, but it assumes that an appropriate image-space contact region has already been identified. In many interactions, obtaining that 2D region is itself ambiguous. A generated person may visually suggest that a hand is lifting a table, but the method must still decide which visible patch of the tabletop or edge corresponds to the hand contact. A person may be shown reaching toward a washing machine, but the correct hand contact should lie near the door or handle region rather than an arbitrary visible surface. A broad object mask is not sufficient in either case. The contact region must be consistent with the instruction, the generated body pose, the body-part identity, the visible object geometry, and the original calibrated scene image.

A related development outside HSI is the use of foundation image models as components of agentic 3D reasoning systems. WorldAgents investigates whether 2D foundation image models can be used as agents for 3D world synthesis, using a multi-agent architecture with a vision-language director, an image generator, and a verifier that evaluates generated frames in both image and reconstruction space [21]. Although its goal is 3D world generation rather than human-scene interaction, it supports a broader observation that is relevant here: foundation image models become more useful for 3D tasks when embedded in iterative generate-and-verify loops rather than used as one-shot image synthesizers.

This motivates the agentic contact-localization stage used in this thesis. The generated human frame is treated as a visual reference for pose, orientation, laterality, and likely contact layout, while the original scene image remains the authoritative canvas. An image model proposes colored overlays for the required scene-side contact regions; the colored pixels are extracted and recomposited onto the unedited original image; and a vision-language evaluator checks whether the resulting composite correctly marks the contact regions implied by the generated interaction. If the contact masks are incorrect, a correction instruction is issued and the process is repeated. The output of this loop is not a final image, but a set of verified body-part-specific 2D contact regions that can be projected to the reconstructed scene mesh.

Several zero-shot human-object interaction methods are related to this direction. DreamHOI uses text-to-image diffusion priors for zero-shot 3D human-object interaction synthesis

with a given human and object, optimizing the human articulation so that the body appears to interact realistically with the object [22]. InterFusion addresses text-driven 3D human-object interaction generation by using text-derived human pose priors and a two-stage generation process to reduce the difficulty of directly producing a full interaction scene from text [23]. InteractAnything uses large language models and diffusion-based object affordance parsing for zero-shot 3D human-object interaction, including extraction of contact points for open-set objects [24]. These works show that foundation models can support interaction synthesis beyond closed object and action datasets.

However, object-centric HOI differs from full-scene HSI. In HOI, the target object is usually explicit, isolated, or provided as an input asset. In HSI, the object is embedded in a larger scene with surrounding geometry, occlusions, support surfaces, and other objects. A body may need to contact both the target object and the floor, avoid nearby obstacles, and remain consistent with the original reconstructed environment. Therefore, scene-side contact grounding in HSI must account not only for object affordance, but also for camera calibration, visible scene geometry, mesh projection, and body-scene collision constraints.

Part-level affordance reasoning in HOI is nevertheless relevant. HOI-PAGE introduces Part Affordance Graphs for zero-shot human-object interaction generation, representing how human body parts engage with object parts and using these relations to guide 4D HOI synthesis [25]. This supports an important trend: realistic interaction generation increasingly requires structured relations between body parts and scene or object parts, rather than only global text prompts or whole-object proximity. The Scene Interaction Graph used in this thesis follows a related motivation, but it is intentionally simpler on the scene side. It specifies which body regions should contact the target object or floor, while the exact scene-side contact region is estimated visually from the generated interaction and original scene image.

Foundation-model priors have also been explored for dynamic HSI. ZeroHSI uses video generation models as motion priors for zero-shot 4D human-scene interaction, reconstructing temporally evolving human-scene interactions through neural human rendering and differentiable optimization [26]. GenHSI similarly uses pretrained video diffusion models for controllable human-scene interaction video generation, decomposing the process into script writing, 3D-aware keyframe previsualization, and animation [27]. These works show that generated videos can provide priors over human motion and interaction dynamics. Their goal, however, is temporal visual synthesis or 4D interaction generation, whereas this thesis focuses on a single static body configuration with explicit metric contact constraints.

Overall, zero-shot interaction generation methods show that foundation models provide useful semantic, visual, and motion priors for HSI and HOI. They reduce dependence on paired 3D interaction datasets and enable broader language control. The remaining challenge is to make these priors geometrically accountable. For static 3D HSI, this means first extracting interaction-specific 2D contact regions on the original scene image, and

then projecting those regions to the reconstructed mesh as scene-side targets for body optimization. The proposed method addresses this gap by using foundation models not only to generate a plausible human interaction image, but also to localize and verify body-part-specific contact regions before enforcing them through SMPL-X optimization.

2.4 Research Gap and Positioning

Prior work has established several important ingredients for human-scene interaction synthesis. Contact-aware 3D HSI methods show that plausible bodies in scenes require contact and collision reasoning. Static synthesis methods show that humans can be generated in 3D environments using learned scene affordances, body-scene compatibility priors, or semantic action-object control. More recent zero-shot methods show that foundation models can provide useful visual and semantic priors for open-vocabulary interactions. However, these directions do not fully address the problem considered in this thesis: synthesizing a static 3D human in a reconstructed scene from an open-vocabulary instruction while grounding each required body contact to a specific scene-side region.

The key missing step is interaction-specific scene-side contact grounding. Object-level grounding is often too coarse, because knowing the target object does not determine where the body should touch it. Part-level grounding is also not always sufficient. Some interactions involve small functional parts, such as handles, controls, rungs, or buttons, but others involve large object parts or broad support surfaces. For example, lifting a table may require hand contact on a plausible region of the tabletop or table edge; sitting on a sofa requires selecting an appropriate patch on the seat or back support; and leaning on a wall requires choosing a contact region consistent with the body pose and orientation. Thus, the desired contact target is not simply an object instance, and sometimes not even an entire semantic object part, but a localized body-part-specific region of the visible scene surface.

Foundation-model-based methods help by producing plausible visual interaction hypotheses, but these hypotheses are not directly usable as 3D constraints. A generated image may suggest that a hand touches an object, but it does not automatically provide a reliable 2D mask for the contacted scene region, nor a corresponding set of 3D mesh vertices. Therefore, zero-shot static HSI requires two linked operations: first, extracting the correct 2D contact regions on the original scene image; and second, projecting those regions into the reconstructed scene as metric contact targets.

This thesis addresses this gap through an agentic contact-grounding pipeline. A Scene Interaction Graph first specifies which body regions should contact the target object or floor. A generated human frame then provides a visual hypothesis for pose, orientation, laterality, and approximate contact layout. Instead of accepting this generated image as

the final output, the method uses it to guide contact localization on the original calibrated scene image. Colored contact overlays are generated, extracted, recomposited onto the unchanged original image, and verified by a vision-language evaluator. This closed-loop process produces body-part-specific 2D contact masks while reducing the effect of scene drift or incorrect one-shot annotations.

The accepted masks are then projected to the reconstructed scene mesh and used as explicit scene-side targets for SMPL-X optimization. The final body is initialized from monocular human reconstruction and refined using contact attraction, penetration avoidance, and initialization priors. This distinguishes the thesis from prior work in three ways: it does not require paired 3D HSI training data, it is not limited to predefined action-object labels, and it treats generated visual evidence as an intermediate source of contact constraints rather than as a final visual result. The contribution is therefore the conversion of open-vocabulary visual interaction evidence into metrically grounded body-part-specific contact targets for static 3D human-scene interaction synthesis.

Chapter 3

Methodology

This chapter describes the proposed zero-shot static human-scene interaction synthesis pipeline. The method converts a natural-language instruction, a calibrated ScanNet++ RGB view, and a reconstructed scene mesh into a single SMPL-X body placed in the original scene coordinate frame.

3.1 Overview

This chapter presents the proposed zero-shot pipeline for static 3D human-scene interaction synthesis. The method takes as input a natural-language interaction instruction y , a calibrated RGB view I from a ScanNet++ scene, the corresponding camera intrinsics K , camera rotation R , camera translation t , and the reconstructed scene mesh M_s . The output is a single SMPL-X body placed in the ScanNet++ scene coordinate frame. The synthesized result is static: it represents the interaction state, or a contact-ready pre-action state, rather than a full motion sequence or an object-state change.

The method is designed to handle both underspecified and detailed interaction instructions. For example, the instruction “a person lifting a table” specifies the interaction at a high level, leaving the exact body configuration and contact layout to be inferred. Functional interactions are more demanding: an instruction such as “a person opening a washing machine” requires the method to infer that contact should occur near the functional door or handle region, not merely somewhere on the washing machine object. The method also supports more detailed instructions, such as “a person on the treadmill with both hands on the side frames and one foot on the belt,” where the user explicitly constrains the pose and the required body-part contacts. In all cases, the central challenge is the same: the language instruction must be converted into metric 3D contact constraints that are consistent with the reconstructed scene.

A visually plausible generated image is not sufficient for this task, because the final body

must also be geometrically compatible with the scene mesh. A generated person may appear reasonable in the input view while still floating above the floor, intersecting furniture, or contacting the wrong object part in 3D. The proposed method therefore combines foundation-model reasoning with explicit geometric grounding. Foundation models are used to infer semantic and visual interaction cues, while camera geometry and optimization convert these cues into explicit 3D contact targets and a physically plausible SMPL-X body.

Figure 3.1 illustrates the complete pipeline. The method consists of five main modules. First, the instruction and scene image are converted into a Scene Interaction Graph (SIG), which identifies the target object or floor, the relevant human body parts, and the required body-part–scene contact edges. Second, an image generation model inserts a plausible human into the original scene view according to the interaction instruction. This generated human frame acts as a visual hypothesis for body pose, orientation, laterality, and approximate contact layout. Third, the generated human frame, the original scene image, and the SIG are used to estimate scene-side contact regions. Each required contact region is marked on the original scene image using a distinct bright color, producing body-part-specific contact masks. When the interaction involves floor support, floor regions are obtained separately using SAM3. The accepted contact masks are then projected to the reconstructed scene mesh to form 3D scene contact targets.

In parallel, the fourth module initializes the human body. The generated human frame is passed to GVHMR, which predicts an initial SMPL-X body. This initialization provides a plausible articulated pose and global body configuration, but it is not assumed to satisfy the scene contact constraints exactly. The fifth module refines the initialized body through scene-aware optimization. During this stage, the scene geometry and projected contact targets are kept fixed, while the SMPL-X translation, global orientation, and body pose are optimized. The objective combines contact attraction, penetration avoidance, and priors that keep the result close to the GVHMR initialization.

The key idea of the proposed approach is that image generation is used not only to synthesize a plausible human interaction image, but also to localize fine-grained scene-side contact regions. These contact annotations are not treated as final outputs. Instead, they are converted into explicit 3D contact targets on the reconstructed mesh and used to guide SMPL-X optimization. In this way, the method combines the flexibility of open-vocabulary foundation models with the geometric accountability required for static 3D human–scene interaction synthesis.

The full prompts used for the foundation-model stages are provided in ?? for reproducibility. The methodology chapter describes the algorithmic role of each model call, while the appendix records the exact prompt wording used by the implementation.

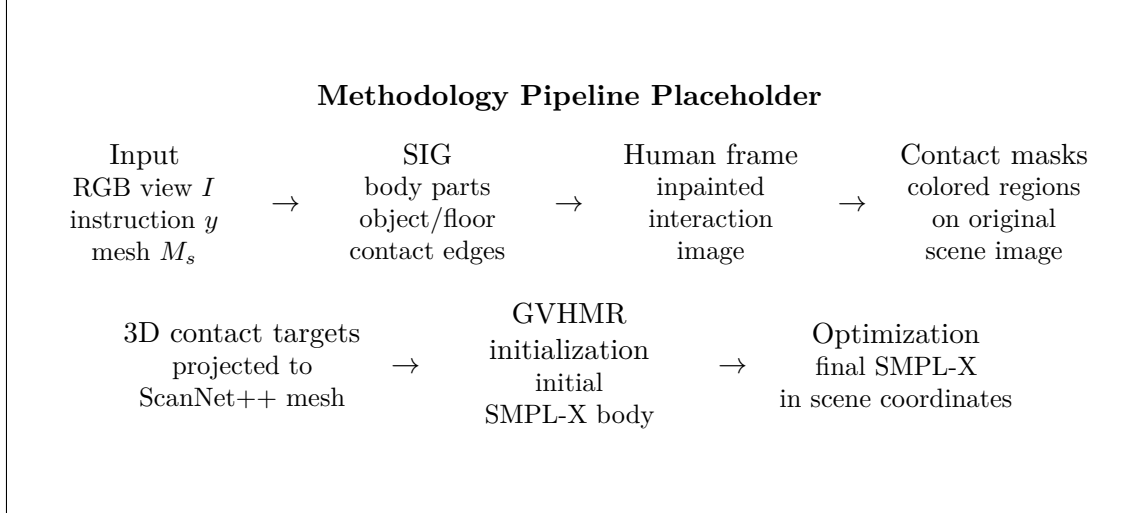


Figure 3.1: Overview of the proposed zero-shot static HSI synthesis pipeline. Given a natural-language instruction, a calibrated ScanNet++ RGB view, and the reconstructed scene mesh, the method generates a Scene Interaction Graph, synthesizes a human interaction image, estimates body-part-specific scene contact regions on the original scene image, projects these contacts to the scene mesh, initializes an SMPL-X body with GVHMR, and refines the body through scene-aware optimization.

3.2 Problem Formulation

Given a natural-language interaction instruction, a calibrated RGB image, and a reconstructed 3D scene, the goal is to synthesize a single human body that realizes the requested interaction in the original scene coordinate frame. Let I denote the selected RGB view from a ScanNet++ scene. The image has spatial resolution

$$I \in \mathbb{R}^{H \times W \times 3}. \quad (3.1)$$

The camera calibration is defined by the intrinsic matrix K , the camera rotation R , and the camera translation t . The reconstructed scene mesh is denoted by

$$M_s = (V_s, F_s), \quad (3.2)$$

where V_s are the scene vertices in the ScanNet++ world coordinate frame and F_s are the triangular faces. The scene vertices satisfy

$$V_s \in \mathbb{R}^{N_s \times 3}. \quad (3.3)$$

The user instruction is denoted by y . It can be underspecified, such as “a person lifting a table,” functional, such as “a person opening a washing machine,” or detailed, such as “a person on the treadmill with both hands on the side frames and one foot on the belt.”

In all cases, the method must infer a body configuration that is consistent with both the language instruction and the reconstructed scene geometry.

The output of the method is a posed SMPL-X body in the same world coordinate frame as the reconstructed scene. SMPL-X represents the body as a differentiable parametric mesh. We denote the body parameters by

$$\theta = (\mathbf{T}, \mathbf{r}, \mathbf{p}, \boldsymbol{\beta}). \quad (3.4)$$

Here, $\mathbf{T}$ is the global translation, $\mathbf{r}$ is the global orientation, $\mathbf{p}$ is the articulated body pose, and $\boldsymbol{\beta}$ are the body shape parameters. Applying the SMPL-X model to these parameters produces the human mesh

$$M_h(\theta) = (V_h(\theta), F_h), \quad (3.5)$$

where $V_h(\theta)$ are the posed body vertices and F_h are the fixed SMPL-X mesh faces. The posed body vertices satisfy

$$V_h(\theta) \in \mathbb{R}^{N_h \times 3}. \quad (3.6)$$

The synthesis target is not a motion sequence. The method estimates one static interaction state, or a contact-ready state, that expresses the instruction. For object-manipulation prompts such as opening a washing machine or lifting a table, the object itself remains fixed in the reconstructed scene. The goal is to synthesize the body configuration that makes the interaction visually and geometrically plausible, not to simulate subsequent object motion or state change.

The instruction y is converted into a set of body–scene contact requirements. These requirements are represented by a Scene Interaction Graph,

$$G = (\mathcal{H}, \mathcal{S}, \mathcal{E}), \quad (3.7)$$

where $\mathcal{H}$ is the set of human body-part nodes, $\mathcal{S}$ is the set of scene nodes, and $\mathcal{E}$ is the set of contact or support edges. Human nodes correspond to supported body parts such as hands, arms, feet, legs, hips, back, and head. Scene nodes correspond either to the target object or to the floor. Each edge is written as

$$e = (h_e, s_e), \quad e \in \mathcal{E}, \quad (3.8)$$

and indicates that body part h_e should contact scene element s_e .

Each human node h_e is associated with a fixed subset of SMPL-X contact vertices. We denote this body-side contact set by

$$B_e \subset \{1, \dots, N_h\}. \quad (3.9)$$

For a given body configuration θ , the corresponding body-side contact points are

$$P_e^h(\theta) = \{V_h(\theta)_i \mid i \in B_e\}. \quad (3.10)$$

The scene-side contact target is not manually annotated. Instead, it is estimated from generated visual contact annotations and projected to the reconstructed scene mesh. For each edge e , this produces a set of scene contact vertices or sampled surface points, denoted by

$$P_e^s \subset \mathbb{R}^3. \quad (3.11)$$

The optimization problem can therefore be stated as finding SMPL-X parameters θ such that the body-side contact points $P_e^h(\theta)$ are close to the corresponding scene-side contact targets P_e^s , while the body remains physically plausible and avoids severe penetration with the scene. The body is initialized using GVHMR from the generated human frame. Let the initial SMPL-X estimate be

$$\theta_0 = (\mathbf{T}_0, \mathbf{r}_0, \mathbf{p}_0, \boldsymbol{\beta}_0). \quad (3.12)$$

During optimization, only the global translation $\mathbf{T}$, global orientation $\mathbf{r}$, and body pose $\mathbf{p}$ are updated. The shape parameters $\boldsymbol{\beta}$ are initialized from GVHMR and kept fixed.

The final objective has the following general form:

$$\mathcal{L}_{\text{total}}(\theta) = \lambda_{\text{contact}} \mathcal{L}_{\text{contact}}(\theta) + \lambda_{\text{penetration}} \mathcal{L}_{\text{penetration}}(\theta) + \lambda_{\text{pose}} \mathcal{L}_{\text{pose}}(\theta) + \lambda_{\text{orient}} \mathcal{L}_{\text{orient}}(\theta). \quad (3.13)$$

The final body parameters are obtained by minimizing this objective with respect to the global translation, global orientation, and body pose:

$$\theta^* = \arg \min_{\mathbf{T}, \mathbf{r}, \mathbf{p}} \mathcal{L}_{\text{total}}(\theta). \quad (3.14)$$

The contact term pulls the specified SMPL-X body regions toward the projected scene contact targets. The penetration term discourages intersections between the human body and the reconstructed scene. The pose and orientation priors keep the optimized body close to the GVHMR initialization so that contact refinement does not produce unnatural body configurations. The following sections describe how the graph G , the human initialization θ_0 , and the scene contact targets P_e^s are obtained.

Table 3.1: Notation used in the methodology chapter.

Symbol	Meaning
I	Calibrated RGB view from a ScanNet++ scene.
K, R, t	Camera intrinsics, rotation, and translation.
$M_s = (V_s, F_s)$	Reconstructed scene mesh in the ScanNet++ world coordinate frame.
y	Natural-language interaction instruction.
$G = (\mathcal{H}, \mathcal{S}, \mathcal{E})$	Scene Interaction Graph.
$e = (h_e, s_e)$	Contact edge between body part h_e and scene element s_e .
B_e	SMPL-X vertex subset assigned to the body part in edge e .
$P_e^h(\theta)$	Body-side contact points for edge e .
P_e^s	Scene-side contact targets projected to the reconstructed scene mesh.
θ	SMPL-X parameters.
θ_0	GVHMR initialization.

3.3 Scene Interaction Graph Generation

The first module converts the open-language instruction into an explicit set of body–scene contact requirements. The purpose of this module is not to solve the full geometry of the interaction, but to determine which human body parts should contact which scene elements. This intermediate representation is called the Scene Interaction Graph (SIG). It provides the symbolic structure that later guides both contact-region estimation and SMPL-X optimization. The exact system prompt used for SIG generation is provided in appendix A.1.1; the main text describes only the functional role of the prompt and the validated graph structure.

The SIG is generated from the user instruction y and the input scene image I . A language or vision-language model is prompted to identify the target object, the human body parts that are physically important for the interaction, and any required floor support. The output is parsed into a graph

$$G = (\mathcal{H}, \mathcal{S}, \mathcal{E}), \quad (3.15)$$

where $\mathcal{H}$ contains human body-part nodes, $\mathcal{S}$ contains scene nodes, and $\mathcal{E}$ contains contact or support edges. The scene nodes are deliberately restricted to the target object and, when required, the floor. The graph does not require explicit object-part labels such as handle, knob, rung, or edge. Instead, it specifies that a particular body part should contact the target object, while the exact functional contact region is estimated later from image-space contact annotations.

The supported human body-part vocabulary is fixed so that each graph node can be mapped to a predefined SMPL-X contact region. The supported parts are left hand, right hand, left arm, right arm, left leg, right leg, left foot, right foot, head, hips, and back.

Table 3.2: Example Scene Interaction Graphs for prompts with different levels of specificity.

Prompt	Scene nodes	Contact/support edges
“A person lifting a table.”	target object: table; floor	left hand $\rightarrow$ table; right hand $\rightarrow$ table; left foot $\rightarrow$ floor; right foot $\rightarrow$ floor.
“A person opening a washing machine.”	target washing machine; floor	right hand $\rightarrow$ washing machine; left foot $\rightarrow$ floor; right foot $\rightarrow$ floor.
“A person on the treadmill with both hands on the side frames and one foot on the belt.”	target treadmill; floor	left hand $\rightarrow$ treadmill; right hand $\rightarrow$ treadmill; left foot $\rightarrow$ treadmill; right foot $\rightarrow$ floor.

This restricted vocabulary prevents the graph from introducing body parts that cannot be grounded to SMPL-X vertices. It also keeps the downstream optimization well defined: every edge in the graph corresponds to one known body-side vertex set and one scene-side target.

The graph edges define the required contacts. For example, an underspecified prompt such as “a person lifting a table” may produce hand–table edges and foot–floor support edges. A functional prompt such as “a person opening a washing machine” may produce a hand–washing-machine edge together with foot–floor support. A detailed prompt such as “a person on the treadmill with both hands on the side frames and one foot on the belt” may produce separate left-hand–treadmill, right-hand–treadmill, left-foot–treadmill, and right-foot–floor edges, depending on the specified pose. These examples illustrate that the graph can represent both inferred contacts and user-specified contacts.

After generation, the SIG is validated before being used by later modules. The validation checks that the graph contains at least one target object or floor node, that each human node belongs to the supported body-part vocabulary, and that every edge references a defined scene node. Duplicate edges and unsupported body parts are removed or rejected. If no usable contact edge remains, the pipeline cannot construct meaningful contact targets and the graph must be regenerated.

Each valid graph edge $e = (h_e, s_e)$ is associated with a body-side contact region on SMPL-X. For a human body part h_e , the corresponding contact region is a fixed vertex subset B_e . These vertex subsets are manually defined from the SMPL-X mesh and represent the surfaces that are most likely to participate in physical contact. For example, hand contacts use palm-side vertices, foot contacts use support-facing foot vertices, and sitting or leaning contacts use appropriate regions on the hips, back, head, or legs. These details are fixed once for the whole method and are reused for all interactions.

The SIG therefore connects language to geometry in a controlled way. It does not determine

the final pose, and it does not localize the exact object surface that should be touched. Instead, it specifies the required body-part–scene relations. The following modules use this graph to generate a visual human interaction hypothesis, estimate the corresponding scene-side contact regions, and optimize the SMPL-X body so that the specified body regions contact the projected scene targets.

3.4 Human Frame Generation

The second module generates a visual hypothesis of the requested interaction in the input scene view. Given the original ScanNet++ image I , the user instruction y , and the Scene Interaction Graph G , an image generation model is prompted to insert a single human into the scene. The output is an edited image in which the generated person appears to perform the requested interaction while the scene remains otherwise unchanged. The full human-frame generation prompt is reported in appendix A.1.2.

The purpose of this module is not to produce the final result. The generated image is used as an intermediate visual prior for two downstream tasks. First, it provides the image evidence needed by GVHMR to initialize an SMPL-X body. Second, it shows the approximate pose, orientation, laterality, and contact layout that the contact-estimation module uses to infer scene-side contact regions. The final output of the method is produced only after contact grounding and 3D SMPL-X optimization.

The image generation prompt is constructed to preserve the original scene as much as possible. The model is instructed to keep the camera viewpoint, room geometry, object layout, lighting, textures, and target object fixed. The prompt also specifies that exactly one person should be inserted. For object-manipulation instructions, the target object should not be moved, opened, lifted, rotated, or deformed by the image model. For example, for an instruction such as “a person opening a washing machine,” the desired output is an image of a person reaching for or contacting the washing-machine door or handle region, while the washing machine itself remains in its original closed state. This is important because the reconstructed scene mesh represents the original object geometry and cannot reflect image-only object edits.

The generated human frame must also respect the body-part requirements encoded in the SIG. If the graph contains hand–object edges, the generated person should place the corresponding hands near the target object. If the graph contains foot–floor edges, the person should appear physically supported by the floor. If the instruction explicitly specifies asymmetric contacts, such as one foot on a treadmill belt and the other foot on the floor, the generated frame should preserve this laterality. These requirements make the generated image a useful visual bridge between language-level interaction intent and later metric optimization.

In practice, the image generation model may produce multiple candidate frames. Let these candidates be denoted by

$$\mathcal{I}_H = \{\hat{I}^{(1)}, \hat{I}^{(2)}, \dots, \hat{I}^{(N)}\}. \quad (3.16)$$

A vision-language evaluator is then used to choose the most suitable candidate from this set. The evaluator compares each candidate against the original image, the interaction instruction, and the expected contact structure. It favors candidates that express the requested interaction clearly, preserve the target object and surrounding scene, contain a plausible full-body pose, and expose the relevant body-part contacts for later reasoning. The prompt used for this candidate-selection step is provided in appendix A.1.3.

The selected human frame is denoted by

$$\hat{I} = \hat{I}^{(j^*)}. \quad (3.17)$$

Minor visual artifacts in the selected frame are acceptable because the generated image is not used as a final rendering. However, candidates with severe object distortions, missing body parts, implausible body scale, incorrect laterality, or unrelated interaction geometry are rejected by the evaluator. This selection step improves the stability of the downstream stages because both GVHMR initialization and contact estimation depend on the generated human being visually consistent with the requested interaction.

The selected human frame $\hat{I}$ remains in the same image coordinate system as the original view I , which allows it to be compared directly with the original scene image in the contact-estimation module. The generated human in $\hat{I}$ is used only as evidence for pose and contact reasoning. The authoritative scene geometry remains the reconstructed ScanNet++ mesh M_s , and the authoritative scene appearance for contact annotation remains the original image I .

3.5 Agentic Diffusion-Based Contact Estimation

The third module estimates the scene-side contact regions required by the interaction. This stage is the main bridge between open-vocabulary visual reasoning and metric 3D optimization. Given the selected human frame $\hat{I}$, the original scene image I , and the Scene Interaction Graph G , the goal is to obtain a set of body-part-specific contact masks on the original scene image. These masks are later projected to the reconstructed scene mesh and used as 3D contact targets.

The key difficulty is that the image model can reason about contacts from the generated human frame, but its edited outputs are not geometrically reliable by themselves. When asked to draw contact overlays, the model may slightly change the target object, modify the background, alter image texture, or shift scene boundaries. The method therefore does not

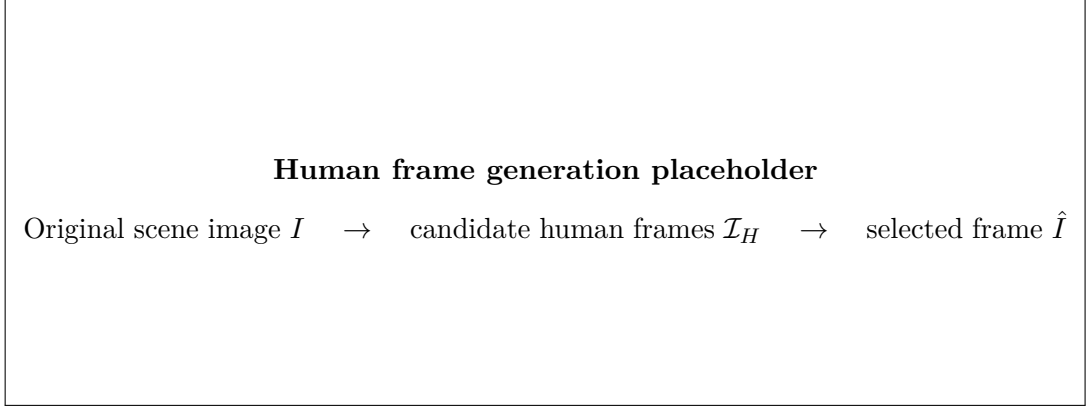


Figure 3.2: Human frame generation and selection. The image model inserts one person into the original ScanNet++ view according to the interaction instruction while preserving the scene, camera viewpoint, and target object. When multiple candidates are generated, a vision-language evaluator selects the frame that best preserves the scene and expresses the required body–scene contacts. The selected frame is used as an intermediate visual hypothesis for GVHMR initialization and contact estimation.

use the generated overlay image as the final annotated image. Instead, it extracts only the solid-color contact regions from the generated output and pastes those regions back onto the unchanged original scene image. The resulting image is called the contact composite. This ensures that the verified masks are always defined in the coordinate system of the calibrated input image I .

3.5.1 Reference, Canvas, and Contact-Overlay Images

For contact estimation, the selected human frame $\hat{I}$ serves as the reference image. It contains the generated person and therefore shows the approximate pose, body laterality, and visible contact configuration. The original ScanNet++ image I serves as the canvas image. It contains the scene without the generated human and preserves the original target-object appearance, location, and boundaries.

For every object-contact edge $e = (h_e, s_e)$ in the SIG, the model must identify the localized scene surface contacted by body part h_e . If the instruction is “a person opening a washing machine,” the corresponding hand contact should be localized near the functional door or handle region. If the instruction specifies a treadmill interaction with both hands on the side frames and one foot on the belt, the contact regions should be localized on the treadmill rails and belt rather than arbitrary treadmill surfaces.

Each required contact edge is assigned a distinct bright color. Let c_e denote the color assigned to edge e . At iteration r , the image model receives the reference image $\hat{I}$, the canvas image I , the target-object description, the current correction instruction if one

exists, and the color mapping from contact edges to colors. It produces an overlay candidate

$$O^{(r)} = \Phi_{\text{con}}(\hat{I}, I, G, \{c_e\}, a^{(r-1)}), \quad (3.18)$$

where Φ_{con} denotes the contact-overlay generation model and $a^{(r-1)}$ is the correction instruction from the previous iteration. For the first iteration, no correction instruction is used. The full contact-overlay prompt is provided in appendix A.1.4.

The intended output $O^{(r)}$ is an edited copy of the canvas image with solid-color masks drawn only on the localized contact regions of the target object. However, because the generated image may also contain unintended changes to the scene, $O^{(r)}$ is treated only as a temporary source from which color masks are extracted.

3.5.2 Overlay Extraction and Composite Construction

After the image model produces the overlay candidate $O^{(r)}$, the method extracts the pixels corresponding to each assigned contact color. For each object-contact edge e , the color extraction step produces a binary mask

$$Z_e^{(r)} = \text{ExtractColor}(O^{(r)}, c_e). \quad (3.19)$$

The extracted mask $Z_e^{(r)}$ contains only pixels whose color matches the assigned body-part color c_e , within a small tolerance. This step discards all other content in the generated overlay image, including any accidental modifications to scene texture, object shape, lighting, or background.

The extracted masks are then placed on top of the original unedited scene image I . This produces the contact composite

$$C^{(r)} = \text{Composite}(I, \{Z_e^{(r)}\}). \quad (3.20)$$

The composite $C^{(r)}$ has the same camera coordinate system, resolution, and scene appearance as the original image I . The only difference is the addition of solid-color contact overlays. This design is important for projection: the final masks must correspond to pixels in the calibrated original image, not to a potentially altered image produced by the generative model.

The composite also separates two roles that would otherwise be entangled. The image generation model is used to infer and draw candidate contact regions, while the original scene image remains the authoritative geometric canvas. Thus, even if the overlay candidate $O^{(r)}$ contains minor scene drift, the downstream verifier and projector only operate on the clean composite $C^{(r)}$.

3.5.3 Agentic Contact Verification and Correction

The extracted contact overlays are verified using a closed-loop vision-language evaluation stage. The evaluator receives three images: the selected human frame $\hat{I}$, the original scene image I , and the contact composite $C^{(r)}$. The selected human frame shows the intended human-object interaction, the original scene image provides the unmodified target-object geometry, and the composite shows the current proposed contact masks on that original scene.

The evaluator checks whether the colored overlays in $C^{(r)}$ correctly mark the scene regions contacted by the generated human in $\hat{I}$. It verifies that all required contacts are present, that no extra contacts are added, that each color corresponds to the correct body part, that each mask lies on the correct object surface, and that each mask has an appropriate size and location. The evaluation prompt is provided in appendix A.1.5.

The evaluator returns either an acceptance decision or a correction instruction. This can be written as

$$(d^{(r)}, a^{(r)}) = \Phi_{\text{eval}}(\hat{I}, I, C^{(r)}, G, \{c_e\}), \quad (3.21)$$

where $d^{(r)}$ is the decision and $a^{(r)}$ is a natural-language correction instruction. If the masks are correct, $d^{(r)}$ is accepted and the loop terminates. If the masks are rejected, the correction instruction is passed back to the contact-overlay generation model for the next iteration. For example, the instruction may ask the model to move a hand mask to the correct handle region, shrink an overly large contact footprint, add a missing contact, remove an extra mask, or correct a body-part color.

This agentic loop is repeated for a limited number of iterations:

$$O^{(r)} \rightarrow \{Z_e^{(r)}\} \rightarrow C^{(r)} \rightarrow (d^{(r)}, a^{(r)}) \rightarrow O^{(r+1)}. \quad (3.22)$$

The first accepted composite is used as the final contact annotation. If no candidate is accepted within the maximum number of attempts, the best available composite is retained only when the contact locations are still usable for optimization; otherwise the interaction is marked as a failure case.

Let r^* denote the selected iteration. The final object-contact masks are

$$M_e^{2D} = Z_e^{(r^*)} \quad \text{for all} \quad e \in \mathcal{E}_{obj}, \quad (3.23)$$

where $\mathcal{E}_{obj}$ is the subset of SIG edges whose scene node is the target object. Small isolated components may be removed after extraction, and masks can be clipped to the visible target-object region when necessary.

3.5.4 Floor Support Masks

Floor contacts are handled separately from object contacts. For support edges such as left-foot-floor or right-foot-floor, the exact pixel location of the foot in the original image is less important than obtaining a geometrically valid support surface. The method therefore uses SAM3 to segment visible floor regions in the original scene image I . The resulting floor mask is projected to the reconstructed scene mesh and used as the scene-side support target for foot contacts.

This separation avoids asking the contact-overlay model to hallucinate precise foot footprints on an empty floor region in the original image. For standing, walking, or balancing interactions, the projected floor region provides a stable support surface, while the SMPL-X optimization determines the final precise foot placement subject to contact, penetration, and pose priors.

Let $\mathcal{E}_{floor}$ denote the subset of SIG edges whose scene node is the floor. For each floor-support edge e , the corresponding 2D support mask is denoted by

$$M_e^{2D} = M_{floor}^{2D} \quad \text{for} \quad e \in \mathcal{E}_{floor}. \quad (3.24)$$

The same floor mask can be reused for multiple foot-support edges, since the distinction between left and right foot is enforced on the body side during optimization.

3.5.5 Projection to 3D Contact Targets

The verified 2D contact masks are projected to the reconstructed scene mesh using the calibrated camera. For a 3D scene point x in the ScanNet++ world coordinate frame, its image projection is

$$\tilde{u} = K(Rx + t), \quad (3.25)$$

followed by perspective division to obtain the pixel coordinate $u = (u_x, u_y)$. A scene vertex or sampled scene point is assigned to contact edge e if its projected pixel lies inside the final mask M_e^{2D} and passes visibility filtering for the selected camera view.

This projection converts each image-space mask into a metric scene-side contact target:

$$P_e^s = \{x \in M_s \mid \pi(x; K, R, t) \in M_e^{2D}\}. \quad (3.26)$$

Here, $\pi(x; K, R, t)$ denotes the calibrated projection from the scene coordinate frame to the image plane. In practice, P_e^s is represented as a set of scene vertices or sampled surface points on the reconstructed mesh. These points remain fixed during optimization and serve as the target regions that the corresponding SMPL-X contact vertices should approach.

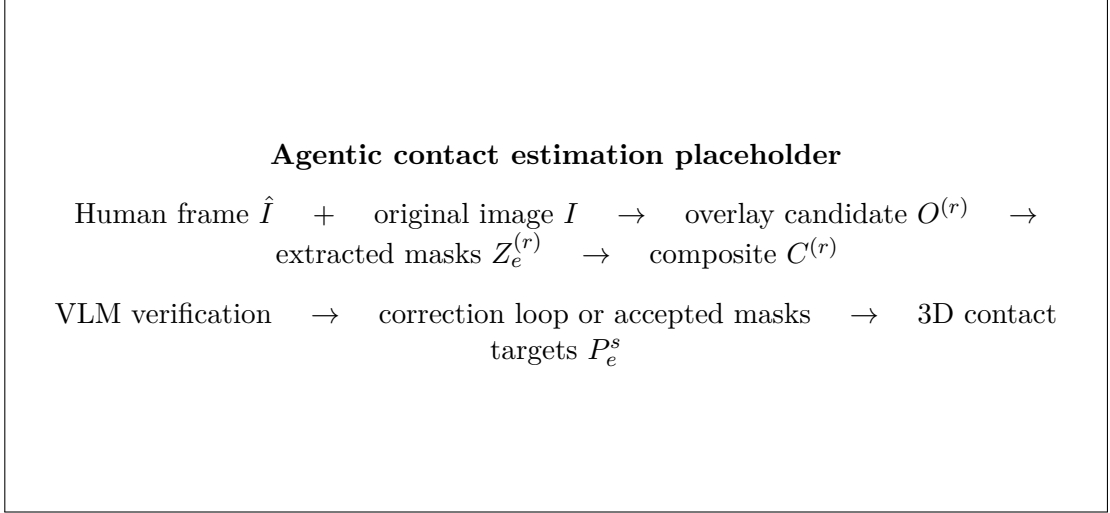


Figure 3.3: Agentic diffusion-based contact estimation. The image model proposes colored contact overlays using the selected human frame as reference and the original scene image as canvas. Since the generated overlay image may still alter the underlying scene appearance or geometry, only the colored mask pixels are extracted and recomposited onto the untouched original image. A vision-language evaluator checks the composite against the human frame and original scene image, then either accepts the masks or returns a correction instruction for another generation attempt. Accepted masks are projected to the reconstructed scene mesh to obtain body-part-specific 3D contact targets.

The final output of the contact-estimation module is the set of projected contact targets

$$\mathcal{P}^s = \{P_e^s \mid e \in \mathcal{E}\}. \quad (3.27)$$

Each element of this set is linked to one SIG edge and therefore to one body-side SMPL-X contact region. The next modules use these targets to initialize and optimize the human body in the reconstructed scene.

3.6 SMPL-X Pose Initialization

The fourth module initializes the human body from the selected generated frame. The contact-estimation stage provides scene-side contact targets, but these targets alone do not determine a complete articulated body. The method therefore uses GVHMR to obtain an initial SMPL-X estimate from the generated human image $\hat{I}$. This initialization provides a plausible full-body pose, body orientation, translation, and shape before scene-aware refinement.

GVHMR is designed for monocular human reconstruction from image or video input. Since the proposed setting uses a single static generated frame, the selected image $\hat{I}$ is converted

into a short static clip by repeating the same frame:

$$\mathcal{V}_{\hat{I}} = (\hat{I}, \hat{I}, \dots, \hat{I}). \quad (3.28)$$

This allows the frame to be processed by the GVHMR pipeline while preserving the static nature of the target interaction. The first reconstructed frame is used as the initialization for the downstream optimization.

Applying GVHMR to the static clip produces an initial SMPL-X estimate in the camera coordinate frame:

$$\theta_0^{cam} = (\mathbf{T}_0^{cam}, \mathbf{r}_0^{cam}, \mathbf{p}_0, \beta_0). \quad (3.29)$$

Here, $\mathbf{T}_0^{cam}$ is the initial global body translation in camera coordinates, $\mathbf{r}_0^{cam}$ is the initial global body orientation, $\mathbf{p}_0$ is the articulated body pose, and β_0 is the estimated SMPL-X body shape. The shape estimate is used as a fixed body identity during optimization.

The final optimization is performed in the ScanNet++ scene coordinate frame, so the GVHMR initialization is transformed from camera coordinates to world coordinates using the selected image camera pose. With the projection convention

$$x^{cam} = Rx^{world} + t, \quad (3.30)$$

the inverse transform maps camera-frame points into the ScanNet++ world frame. The initial translation is converted as

$$\mathbf{T}_0 = R^\top (\mathbf{T}_0^{cam} - t). \quad (3.31)$$

The global body orientation is converted by applying the inverse camera rotation to the camera-frame body orientation. If $\mathbf{R}_0^{cam}$ denotes the rotation matrix corresponding to $\mathbf{r}_0^{cam}$, the world-frame orientation is

$$\mathbf{R}_0 = R^\top \mathbf{R}_0^{cam}. \quad (3.32)$$

The corresponding axis-angle representation is denoted by $\mathbf{r}_0$.

The resulting initialization in the scene coordinate frame is

$$\theta_0 = (\mathbf{T}_0, \mathbf{r}_0, \mathbf{p}_0, \beta_0). \quad (3.33)$$

This initialization is visually plausible because it is estimated from the generated human frame, but it is not guaranteed to satisfy the projected contact targets or avoid collisions with the reconstructed scene mesh. In particular, the generated image may contain small inconsistencies in scale, depth, contact location, or body placement. The role of the next stage is therefore to refine $\mathbf{T}_0$, $\mathbf{r}_0$, and $\mathbf{p}_0$ using the scene geometry and the contact targets obtained from the previous module.

During optimization, the shape parameters β_0 are kept fixed. This avoids using contact refinement to compensate for geometric errors by changing body identity. The optimized variables are limited to global translation, global orientation, and articulated body pose, while the reconstructed scene mesh and projected contact targets remain fixed.

3.7 Scene-Aware SMPL-X Optimization

The final module refines the initialized SMPL-X body using the reconstructed scene geometry and the projected contact targets. The GVHMR estimate provides a plausible human pose from the generated image, but it is not guaranteed to satisfy the contact requirements in 3D. In particular, the body may be slightly offset from the target object, may contact an incorrect surface, or may intersect the reconstructed scene. The optimization stage corrects these errors while preserving the plausibility of the initial body pose.

3.7.1 Optimized Variables

The optimization starts from the world-frame initialization

$$\theta_0 = (\mathbf{T}_0, \mathbf{r}_0, \mathbf{p}_0, \beta_0). \quad (3.34)$$

Only the global translation $\mathbf{T}$, global orientation $\mathbf{r}$, and articulated body pose $\mathbf{p}$ are optimized. The body shape β_0 is kept fixed throughout the optimization. The optimized body parameters are therefore

$$\theta = (\mathbf{T}, \mathbf{r}, \mathbf{p}, \beta_0). \quad (3.35)$$

Keeping the shape fixed prevents the optimizer from satisfying contact constraints by changing the body identity or body proportions. The optimization is instead forced to explain the interaction through pose and placement.

For any current parameter state θ , the SMPL-X model produces a posed human mesh

$$M_h(\theta) = (V_h(\theta), F_h). \quad (3.36)$$

The scene mesh M_s and the projected contact targets P_e^s remain fixed. Thus, the optimization changes only the human body, not the reconstructed scene or the estimated scene-side contact regions.

3.7.2 Contact Objective

The contact objective attracts the specified SMPL-X body regions toward the corresponding projected scene contact targets. For each SIG edge e , the body-side contact vertices are

given by the fixed vertex subset B_e , and the corresponding scene-side target is P_e^s . The body-side contact points under the current pose are

$$P_e^h(\theta) = \{V_h(\theta)_i \mid i \in B_e\}. \quad (3.37)$$

The contact loss is defined as a nearest-neighbor distance from the selected body contact vertices to the projected scene contact targets:

$$\mathcal{L}_{\text{contact}}(\theta) = \sum_{e \in \mathcal{E}} \frac{1}{|B_e|} \sum_{i \in B_e} \min_{q \in P_e^s} \|V_h(\theta)_i - q\|_2^2. \quad (3.38)$$

This loss is edge-specific: hand vertices are pulled toward hand contact targets, foot vertices toward floor or object support targets, and so on. The SIG therefore determines which body region should be attracted to which scene region, while the projected masks determine the metric 3D location of the target.

The one-sided nearest-neighbor formulation is useful because the scene target can contain multiple vertices or sampled surface points. The body contact region is encouraged to approach the nearest compatible part of the target region without requiring a fixed point-to-point correspondence between SMPL-X vertices and scene vertices.

3.7.3 Penetration Objective

Contact attraction alone can produce physically invalid solutions if the body moves into the scene geometry. To discourage intersections, the optimization includes a penetration objective based on a signed distance representation of the SMPL-X body. The method uses VolumetricSMPL to query whether sampled scene points lie inside the current posed body.

Let $\mathcal{Q}_s$ denote a set of scene points sampled from the reconstructed mesh near the current body. Let $\phi_\theta(x)$ be the signed distance from a scene point x to the posed SMPL-X body, with negative values indicating that x lies inside the body volume. The penetration loss is

$$\mathcal{L}_{\text{penetration}}(\theta) = \frac{1}{|\mathcal{Q}_s|} \sum_{x \in \mathcal{Q}_s} [\max(0, -\phi_\theta(x))]^2. \quad (3.39)$$

This term penalizes scene points that are inside the human body and assigns zero cost to points outside the body. It therefore discourages severe body-scene interpenetration while still allowing nearby contact.

3.7.4 Pose and Orientation Priors

The contact and penetration losses are not sufficient to guarantee a natural human pose. Since the initialization from GVHMR is visually plausible, the optimization includes priors

that keep the refined body close to the initial estimate. The pose prior penalizes large deviations from the initial articulated pose:

$$\mathcal{L}_{\text{pose}}(\theta) = \|\mathbf{p} - \mathbf{p}_0\|_2^2. \quad (3.40)$$

The orientation prior penalizes large changes in the global body orientation. Let $\mathbf{R}(\mathbf{r})$ and $\mathbf{R}(\mathbf{r}_0)$ be the rotation matrices corresponding to the current and initial global orientations. The orientation prior is defined using the geodesic distance on rotations:

$$\mathcal{L}_{\text{orient}}(\theta) = \left\| \text{Log} \left(\mathbf{R}(\mathbf{r}_0)^\top \mathbf{R}(\mathbf{r}) \right) \right\|_2^2. \quad (3.41)$$

Here, Log maps a rotation matrix to its axis-angle vector. This term prevents the optimizer from satisfying contacts through unrealistic global rotations that are inconsistent with the generated human frame.

3.7.5 Total Objective

The final optimization objective combines contact attraction, penetration avoidance, and the initialization priors:

$$\mathcal{L}(\theta) = \lambda_{\text{contact}} \mathcal{L}_{\text{contact}}(\theta) + \lambda_{\text{penetration}} \mathcal{L}_{\text{penetration}}(\theta) + \lambda_{\text{pose}} \mathcal{L}_{\text{pose}}(\theta) + \lambda_{\text{orient}} \mathcal{L}_{\text{orient}}(\theta). \quad (3.42)$$

The optimized parameters are obtained by

$$\theta^* = \arg \min_{\mathbf{T}, \mathbf{r}, \mathbf{p}} \mathcal{L}(\theta), \quad \boldsymbol{\beta} = \boldsymbol{\beta}_0. \quad (3.43)$$

The contact term enforces the functional body–scene relationships inferred by the previous modules. The penetration term prevents the body from achieving contact by moving through scene geometry. The pose and orientation priors preserve the plausibility of the GVHMR initialization. Together, these terms refine the initial SMPL-X estimate into a geometrically grounded static interaction state.

3.7.6 Optimization Procedure

The optimization is initialized with θ_0 from GVHMR. At each iteration, the SMPL-X mesh is recomputed, body contact vertices are matched to their corresponding scene contact targets, penetration is evaluated against nearby scene samples, and gradients are backpropagated through the differentiable SMPL-X model. The scene mesh, projected contact targets, and body shape remain fixed.

The optimization returns the final parameters

$$\theta^* = (\mathbf{T}^*, \mathbf{r}^*, \mathbf{p}^*, \beta_0). \quad (3.44)$$

The final output mesh is

$$M_h^* = M_h(\theta^*), \quad (3.45)$$

which is already expressed in the ScanNet++ scene coordinate frame. This mesh is the final synthesized static human–scene interaction result.

The optimization is carried out using Adam for a fixed budget of 2000 iterations. This fixed optimization budget is used across all interactions.

Chapter 4

Experiments

This chapter defines the experimental protocol used to validate the thesis claims.

4.1 Research Questions

State the questions the experiments answer. Each question should map to a table, figure, ablation, or qualitative analysis.

4.2 Datasets

Describe all datasets, splits, annotations, preprocessing steps, and any generated data.

4.3 Baselines

List baseline methods and explain how they were configured. Make comparisons fair and reproducible.

4.4 Metrics

Define quantitative metrics and explain what each metric captures. Include limitations of each metric where relevant.

Table 4.1: Example experiment configuration table. Replace with the final setup.

Item	Value
Dataset split	train / validation / test
Batch size	–
Optimizer	–
Hardware	–

4.5 Experimental Setup

Specify training schedule, optimizer, learning rate, batch size, number of runs, random seeds, hardware, and software versions.

Chapter 5

Results and Discussion

This chapter reports the main results, ablations, qualitative examples, and limitations.

5.1 Main Results

Present the primary quantitative results first. Explain the trend, not just the numbers.

5.2 Ablation Studies

Isolate the effect of each contribution. Keep ablations tied to the claims made in the introduction.

5.3 Qualitative Results

Show representative successes and failures. Qualitative figures should make the method’s behavior inspectable.

5.4 Limitations

Discuss failure cases, data bias, compute cost, assumptions, and evaluation gaps. Strong limitations make the thesis more credible.

Table 5.1: Main result table placeholder.

Method	Metric 1	Metric 2	Metric 3
Baseline	–	–	–
Proposed	–	–	–

5.5 Discussion

Connect the results back to the motivation. Explain what the experiments imply for the broader research problem.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

Summarize the thesis in a few paragraphs: problem, approach, evidence, and main takeaway. Avoid introducing new experiments here.

6.2 Future Work

Describe concrete extensions that follow from the limitations and results. Separate near-term engineering improvements from longer-term research directions.

6.3 Closing Remarks

End with a brief statement about the expected impact or usefulness of the work.

Bibliography

- [1] Angela Dai, Angel X. Chang, Manolis Savva, Maciej Halber, Thomas Funkhouser, and Matthias Nießner. ScanNet: Richly-annotated 3d reconstructions of indoor scenes. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017.
- [2] Chandan Yeshwanth, Yueh-Cheng Liu, Matthias Nießner, and Angela Dai. ScanNet++: A high-fidelity dataset of 3d indoor scenes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [3] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. Expressive body capture: 3d hands, face, and body from a single image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.
- [4] Mohamed Hassan, Vasileios Choutas, Dimitrios Tzionas, and Michael J. Black. Resolving 3d human pose ambiguities with 3d scene constraints. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019.
- [5] Siwei Zhang, Yan Zhang, Qianli Ma, Michael J. Black, and Siyu Tang. PLACE: Proximity learning of articulation and contact in 3d environments. In *International Conference on 3D Vision*, pages 642–651, 2020.
- [6] Mohamed Hassan, Partha Ghosh, Joachim Tesch, Dimitrios Tzionas, and Michael J. Black. Populating 3d scenes by learning human-scene interaction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021.
- [7] Chun-Hao P. Huang, Hongwei Yi, Markus Höschle, Matvey Safroshkin, Tsvetelina Alexiadis, Senya Polikovsky, Daniel Scharstein, and Michael J. Black. Capturing and inferring dense full-body human-scene contact. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- [8] Shashank Tripathi, Agniv Chatterjee, Jean-Claude Passy, Hongwei Yi, Dimitrios Tzionas, and Michael J. Black. DECO: Dense estimation of 3d human-scene contact in the wild. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

- [9] Omid Taheri, Nima Ghorbani, Michael J. Black, and Dimitrios Tzionas. GRAB: A dataset of whole-body human grasping of objects. In *European Conference on Computer Vision*, 2020.
- [10] Bharat Lal Bhatnagar, Xianghui Xie, Ilya A. Petrov, Cristian Sminchisescu, Christian Theobalt, and Gerard Pons-Moll. BEHAVE: Dataset and method for tracking human object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- [11] Xueting Li, Sifei Liu, Kihwan Kim, Xiaolong Wang, Ming-Hsuan Yang, and Jan Kautz. Putting humans in a scene: Learning affordance in 3d indoor environments. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12368–12376, 2019.
- [12] Yan Zhang, Mohamed Hassan, Heiko Neumann, Michael J. Black, and Siyu Tang. Generating 3d people in scenes without people. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6194–6204, 2020.
- [13] Kaifeng Zhao, Shaofei Wang, Yan Zhang, Thabo Beeler, and Siyu Tang. Compositional human-scene interaction synthesis with semantic control. In *European Conference on Computer Vision*, 2022.
- [14] Zan Wang, Yixin Chen, Tengyu Liu, Yixin Zhu, Wei Liang, and Siyuan Huang. HUMANISE: Language-conditioned human motion generation in 3d scenes. In *Advances in Neural Information Processing Systems*, 2022.
- [15] Zhi Cen, Huaijin Pi, Sida Peng, Zehong Shen, Minghui Yang, Shuai Zhu, Hujun Bao, and Xiaowei Zhou. Generating human motion in 3d scenes from text descriptions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [16] Hongwei Yi, Justus Thies, Michael J. Black, Xue Bin Peng, and Davis Rempe. Generating human interaction motions in scenes with text control. In *European Conference on Computer Vision*, 2024.
- [17] Siyuan Huang, Zan Wang, Puhao Li, Baoxiong Jia, Tengyu Liu, Yixin Zhu, Wei Liang, and Song-Chun Zhu. Diffusion-based generation, optimization, and planning in 3d scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16750–16761, 2023.
- [18] Nan Jiang, Zhiyuan Zhang, Hongjie Li, Xiaoxuan Ma, Zan Wang, Yixin Chen, Tengyu Liu, Yixin Zhu, and Siyuan Huang. Scaling up dynamic human-scene interaction modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1737–1747, 2024.

- [19] Lei Li and Angela Dai. GenZI: Zero-shot 3d human-scene interaction generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [20] Jie Liu, Yu Sun, Alpar Cseke, Yao Feng, Nicolas Heron, Michael J. Black, and Yan Zhang. Open-vocabulary functional 3d human-scene interaction generation, 2026.
- [21] Ziya Erkoç, Angela Dai, and Matthias Nießner. Worldagents: Can foundation image models be agents for 3d world models?, 2026.
- [22] Thomas Hanwen Zhu, Ruining Li, and Tomas Jakab. DreamHOI: Subject-driven generation of 3d human-object interactions with diffusion priors, 2024.
- [23] Sisi Dai, Wenhao Zhang, Haowen Li, Peng Wang, Yongwei Zhang, Kai Wang, Shouhong Wang, Tieniu Tan, Xiaogang Liu, and Xiaogang Wang. Text-driven generation of 3d human-object interaction. In *European Conference on Computer Vision*, 2024.
- [24] Jinlu Zhang, Yixin Chen, Zan Wang, Jie Yang, Yizhou Wang, and Siyuan Huang. InteractAnything: Zero-shot human object interaction synthesis via LLM feedback and object affordance parsing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [25] Lei Li and Angela Dai. HOI-PAGE: Zero-shot human-object interaction generation with part affordance guidance, 2025.
- [26] Hongjie Li, Hong-Xing Yu, Jiaman Li, and Jiajun Wu. ZeroHSI: Zero-shot 4d human-scene interaction by video generation, 2024.
- [27] Zekun Li, Rui Zhou, Rahul Sajnani, Xiaoyan Cong, Daniel Ritchie, and Srinath Sridhar. GenHSI: Controllable generation of human-scene interaction videos, 2025.
- [28] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.
- [29] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A skinned multi-person linear model. *ACM Transactions on Graphics*, 34(6):248:1–248:16, 2015.
- [30] Federica Bogo, Angjoo Kanazawa, Christoph Lassner, Peter Gehler, Javier Romero, and Michael J. Black. Keep it SMPL: Automatic estimation of 3d human pose and shape from a single image. In *European Conference on Computer Vision*, 2016.
- [31] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Doll’ar, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4015–4026, 2023.

- [32] Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Didac Suris, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, Jie Lei, Tengyu Ma, Baishan Guo, Arpit Kalla, Markus Marks, et al. SAM 3: Segment anything with concepts, 2025.
- [33] Zehong Shen, Huaijin Pi, Yan Xia, Zhi Cen, Sida Peng, Zechen Hu, Hujun Bao, Ruizhen Hu, and Xiaowei Zhou. World-grounded human motion recovery via gravity-view coordinates. In *SIGGRAPH Asia Conference Proceedings*, 2024.
- [34] Manolis Savva, Angel X. Chang, Pat Hanrahan, Matthew Fisher, and Matthias Nießner. PiGraphs: Learning interaction snapshots from observations. *ACM Transactions on Graphics*, 35(4), 2016.
- [35] Ilya A. Petrov, Riccardo Marin, Julian Chibane, and Gerard Pons-Moll. TriDi: Trilateral diffusion of 3d humans, objects, and interactions. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025.
- [36] Yinghao Huang, Omid Taheri, Michael J. Black, and Dimitrios Tzionas. InterCap: Joint markerless 3d tracking of humans and objects in interaction from multi-view RGB-D images. *International Journal of Computer Vision*, 2024.
- [37] Xiaohan Zhang, Bharat Lal Bhatnagar, Vladimir Guzov, Sebastian Starke, and Gerard Pons-Moll. COUCH: Towards controllable human-chair interactions. In *European Conference on Computer Vision*, 2022.
- [38] Nan Jiang, Tengyu Liu, Zhexuan Cao, Jieming Cui, Zhiyuan Zhang, Yixin Chen, He Wang, Yixin Zhu, and Siyuan Huang. Full-body articulated human-object interaction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [39] Jiaman Li, Jiajun Wu, and C. Karen Liu. Object motion guided human motion synthesis. *ACM Transactions on Graphics*, 42(6):1–11, 2023.
- [40] Jiashun Wang, Huazhe Xu, Jingwei Xu, Sifei Liu, and Xiaolong Wang. Synthesizing long-term 3d human motion and interaction in 3d scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021.
- [41] Mohamed Hassan, Duygu Ceylan, Ruben Villegas, Jun Saito, Jimei Yang, Yi Zhou, and Michael J. Black. Stochastic scene-aware motion prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021.
- [42] Yan Zhang and Siyu Tang. The wanderings of odysseus in 3d scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20481–20491, 2022.

- [43] Jingbo Wang, Yu Rong, Jingyuan Liu, Sijie Yan, Dahua Lin, and Bo Dai. Towards diverse and natural scene-aware 3d human motion synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20460–20469, 2022.
- [44] Kaifeng Zhao, Yan Zhang, Shaofei Wang, Thabo Beeler, and Siyu Tang. Synthesizing diverse human motions in 3d indoor scenes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 14738–14749, 2023.
- [45] Zan Wang, Yixin Chen, Baoxiong Jia, Puhao Li, Jinlu Zhang, Jingze Zhang, Tengyu Liu, Yixin Zhu, Wei Liang, and Siyuan Huang. Move as you say, interact as you can: Language-guided human motion generation with scene affordance. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 433–444, 2024.
- [46] Inwoo Hwang, Bing Zhou, Young Min Kim, Jian Wang, and Chuan Guo. SceneMI: Motion in-betweening for modeling human-scene interactions, 2025.
- [47] Anindita Ghosh, Vladislav Golyanik, Taku Komura, Philipp Slusallek, Christian Theobalt, and Rishabh Dabral. SceMoS: Scene-aware 3d human motion synthesis by planning with geometry-grounded tokens. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- [48] Pradyumna Yalandur Muralidhar, Yuxuan Xue, Xianghui Xie, Margaret Kostyrko, and Gerard Pons-Moll. PhysIC: Physically plausible 3d human-scene interaction and contact from a single image. In *SIGGRAPH Asia 2025 Conference Papers*, 2025.
- [49] Sifan Ye, Yixing Wang, Jiaman Li, Dennis Park, C. Karen Liu, Huazhe Xu, and Jiajun Wu. Scene synthesis from human motion. In *SIGGRAPH Asia 2022 Conference Papers*, 2022.
- [50] Xiaogang Peng, Yiming Xie, Zizhao Wu, Varun Jampani, Deqing Sun, and Huaizu Jiang. HOI-Diff: Text-driven synthesis of 3d human-object interactions using diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2025.
- [51] Sirui Xu, Zhengyuan Li, Yu-Xiong Wang, and Liang-Yan Gui. InterDiff: Generating 3d human-object interactions with physics-informed diffusion. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [52] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022.

- [53] Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. RePaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- [54] Tim Brooks, Aleksander Holynski, and Alexei A. Efros. InstructPix2Pix: Learning to follow image editing instructions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [55] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3836–3847, 2023.
- [56] OpenAI. Image generation. <https://developers.openai.com/api/docs/guides/image-generation>, 2025. Accessed: 2026-06-28.
- [57] Google. Nano Banana 2: Gemini ai image generator and photo editor. <https://gemini.google/overview/image-generation/>, 2026. Accessed: 2026-06-28.
- [58] Marko Mihajlovic, Siwei Zhang, Gen Li, Kaifeng Zhao, Lea Müller, and Siyu Tang. VolumetricSMPL: A neural volumetric body model for efficient interactions, contacts, and collisions. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025.
- [59] Yang Zheng, Yanchao Yang, Kaichun Mo, Jiaman Li, Tao Yu, Yebin Liu, C. Karen Liu, and Leonidas J. Guibas. GIMO: Gaze-informed human motion prediction in context. In *European Conference on Computer Vision*, 2022.
- [60] Siwei Zhang, Qianli Ma, Yan Zhang, Zhiyin Qian, Taein Kwon, Marc Pollefeys, Federica Bogo, and Siyu Tang. EgoBody: Human body shape and motion of interacting people from head-mounted devices. In *European Conference on Computer Vision*, 2022.
- [61] Lu Ling, Chen-Hsuan Lin, Tsung-Yi Lin, Yifan Ding, Yu Zeng, Yichen Sheng, Yunhao Ge, Ming-Yu Liu, Aniket Bera, and Zhaoshuo Li. Scenethesis: A language and vision agentic framework for 3d scene generation, 2025.

Appendix A

Additional Material

A.1 Foundation Model Prompts

This appendix records the system prompts used by the foundation-model stages in chapter 3. The symbols below are used in the methodology chapter.

Table A.1: System-prompt symbols used in the methodology.

Symbol	Stage	Source or status
Π_{SIG}	SIG generation	4DHSI/01_Generate_SIG/system_prompt_sig.md
Π_{H}	Human-frame inpainting	4DHSI/02_Generate_Human_Frame/system_prompt_human_inpaint.md
Π_{sel}	Best human-frame selection	Placeholder; final system prompt not fixed yet.
Π_{con}	Contact overlay generation	4DHSI/03_Estimate_Contact_Agentic/system_prompt_estimate_contact_agentic.md
Π_{eval}	Contact VLM evaluation	4DHSI/03_Estimate_Contact_Agentic/vlm_prompt_evaluate_contact.md

A.1.1 SIG Generation System Prompt Π_{SIG}

You generate a Scene Interaction Graph (SIG) for a static human-scene interaction.

Produce a compact graph containing the target object or objects, relevant human body parts, direct physical contact edges, and a static interaction description.

Input:

- A scene image from the selected camera view.
- A JSON request:

```
{
  "interaction": "short interaction description"
}
```

Output only valid JSON. Do not include markdown or commentary.

Required output schema:

```
{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "short primary target object name"
    },
    {
      "id": "target_object_2",
      "label": "short secondary target object name"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "brief reason for this contact"
    }
  ],
  "interaction": "Static interaction description under 150 words."
}
```

Rules:

1. Use the interaction text to identify the intended human-scene contact, and use the
↪ image only to understand visible object context.
2. `target\_objects` should contain the scene object that receive direct human contact
↪ in the static pose.
3. Use one target object when the contact is with a single object or multiple
↪ structural parts of the same object, such as a chair back, bed frame, or bathtub
↪ side.
4. Use two target objects only when two separate scene objects are directly contacted.
↪ For example, use "bathtub" and "curtain rail" when one foot contacts the bathtub
↪ and one hand contacts the curtain rail.
5. Keep target object labels short noun phrases.
6. Use human part nodes only from this vocabulary:
 - left hand
 - right hand
 - left arm
 - right arm
 - left leg
 - right leg
 - left foot
 - right foot
 - head
 - hips
 - back

7. Include a human part node only when that part is physically important for contact
 ↳ or pose.
8. `scene\_nodes` can include only:
 - target_object_1
 - target_object_2
 - floor
9. Include `target\_object\_2` only if it exists in `target\_objects` and has at least
 ↳ one direct contact edge.
10. `interaction\_edges` represent direct physical contact only.
11. Use `scene\_element: "target\_object\_1"` or `scene\_element: "target\_object\_2"` for
 ↳ contacts with the corresponding target object.
12. Use floor edges only for required direct floor support. If there is no floor
 ↳ contact, simply omit `floor` from `scene\_nodes` and `interaction\_edges`.
13. Do not create object-part nodes.
14. Keep interaction edges focused on physically necessary contacts.
15. The output `interaction` must describe the static pose and physical contacts under
 ↳ 150 words. It must explicitly restate every `interaction\_edges` contact using the
 ↳ same side-specific human part names, such as "left hand", "right foot", or "back",
 ↳ and the contacted scene object label or floor. Avoid vague phrases such as "one
 ↳ hand", "both feet", or "legs" when the edge uses specific left/right parts. It
 ↳ should also describe clothing, shoes, hair, and facial expression.

Few-shot examples:

These few-shot examples are written without a scene image. They demonstrate the expected JSON structure and explicit contact wording. For the real request, use the provided scene image to decide the exact contact configuration, including which side-specific hand, foot, leg, or arm is in contact.

Example 1 input JSON:

```
{
  "interaction": "a person lifting the brown small wooden table with both hands"
}
```

Example 1 output:

```
{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "brown small wooden table"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand",
    "person 1, left foot",
    "person 1, right foot"
  ],
  "scene_nodes": [
    "target_object_1",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand grips the table while preparing to lift"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_1",

```



```

    "notes": "the right hand grips the opposite side of the table"
  },
  {
    "human_part": "left foot",
    "scene_element": "floor",
    "notes": "the left foot supports the standing pose on the floor"
  },
  {
    "human_part": "right foot",
    "scene_element": "floor",
    "notes": "the right foot supports the standing pose on the floor"
  }
],
"interaction": "A person stands beside the brown small wooden table in a
↪ forward-leaning pose. The left hand grips one side of the brown small wooden
↪ table, the right hand grips the opposite side, the left foot is planted on the
↪ floor, and the right foot is planted on the floor. The person has short black
↪ hair, a neutral focused facial expression, a gray shirt, blue jeans, and white
↪ sneakers."
}

```

Example 2 input JSON:

```

{
  "interaction": "a person lying on bed with his head on the pillow"
}

```

Example 2 output:

```

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "bed with pillows"
    }
  ],
  "human_part_nodes": [
    "person 1, left leg",
    "person 1, right leg",
    "person 1, head",
    "person 1, hips",
    "person 1, back"
  ],
  "scene_nodes": [
    "target_object_1"
  ],
  "interaction_edges": [
    {
      "human_part": "head",
      "scene_element": "target_object_1",
      "notes": "the head rests on the pillow as part of the bed setup"
    },
    {
      "human_part": "hips",
      "scene_element": "target_object_1",
      "notes": "the hips are supported by the mattress"
    },
    {
      "human_part": "back",
      "scene_element": "target_object_1",
      "notes": "the back rests against the mattress"
    }
  ],
  {

```

```

    "human_part": "left leg",
    "scene_element": "target_object_1",
    "notes": "the left leg rests on the bed"
  },
  {
    "human_part": "right leg",
    "scene_element": "target_object_1",
    "notes": "the right leg rests on the bed"
  }
],
"interaction": "A person lies on the bed with pillows in a relaxed static pose. The
↪ head rests on a pillow, the back rests on the mattress, the hips are supported
↪ by the mattress, the left leg rests on the bed, and the right leg rests on the
↪ bed. The person has short black hair, a calm sleepy facial expression, a gray
↪ shirt, soft pants, and socks."
}

```

Example 3 input JSON:

```

{
  "interaction": "a person stepping over the bathtub wall, one foot inside the tub,
  ↪ the other foot still on the floor, one hand holding the curtain rail for balance"
}

```

Example 3 output:

```

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "bathtub"
    },
    {
      "id": "target_object_2",
      "label": "curtain rail"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, left foot",
    "person 1, right foot"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_2",
      "notes": "the left hand grips the curtain rail for balance"
    },
    {
      "human_part": "left foot",
      "scene_element": "target_object_1",
      "notes": "the left foot presses inside the bathtub while stepping over the wall"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot stays planted on the floor outside the bathtub"
    }
  ]
}

```

```

],
"interaction": "A person steps over the bathtub wall in a careful balancing pose.
↳ The left hand grips the curtain rail, the left foot presses inside the bathtub,
↳ and the right foot is planted on the floor outside the tub. The person has short
↳ black hair, a focused facial expression, a gray shirt, rolled-up pants, and bare
↳ feet."
}

```

Example 4 input JSON:

```

{
  "interaction": "a person bending down in front of the lower drawer, one hand pulling
↳ the cabinet door open and the other hand holding the counter edge for balance"
}

```

Example 4 output:

```

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "lower cabinet door"
    },
    {
      "id": "target_object_2",
      "label": "counter edge"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand",
    "person 1, left foot",
    "person 1, right foot",
    "person 1, hips"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand pulls the lower cabinet door"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_2",
      "notes": "the right hand holds the counter edge for balance"
    },
    {
      "human_part": "left foot",
      "scene_element": "floor",
      "notes": "the left foot supports the squat on the floor"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot supports the squat on the floor"
    }
  ],
}

```

```

"interaction": "A person bends down in front of the lower cabinet door in a balanced
↪ pose. The left hand pulls the lower cabinet door, the right hand holds the
↪ counter edge, the left foot is planted on the floor, and the right foot is
↪ planted on the floor. The person has short black hair, a neutral concentrated
↪ facial expression, a gray shirt, blue jeans, and white sneakers."
}

```

Example 5 input JSON:

```

{
  "interaction": "a person hanging from the pull-up bar with both hands, elbows bent,
↪ knees tucked upward, and both feet off the floor"
}

```

Example 5 output:

```

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "pull-up bar"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand"
  ],
  "scene_nodes": [
    "target_object_1"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand grips the pull-up bar"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_1",
      "notes": "the right hand grips the pull-up bar"
    }
  ],
  "interaction": "A person hangs from the pull-up bar with elbows bent and knees
↪ tucked upward. The left hand grips the pull-up bar, the right hand grips the
↪ pull-up bar, the left foot stays off the floor, and the right foot stays off the
↪ floor. The person has short black hair, a strained focused facial expression, a
↪ gray athletic shirt, dark shorts, and sneakers."
}

```

A.1.2 Human-Frame Inpainting System Prompt Π_H

Input Image:

- The image is the original scene. Use the interaction description to identify the
↪ existing
target object that must remain fixed and contacted.

Scene Preservation Instructions:

- Use the provided scene image as a fixed camera reference.

- Insert exactly one person performing the described interaction with the target
↪ object.
- This frame must represent the pre-motion start state: the person is already in
↪ natural contact

with the target object of the scene, but no visible object motion has begun yet.

- Target object is already present in the scene. Do not add it.
- Preserve the camera pose, field of view, perspective, room geometry, lighting,
↪ shadows, textures,
and all existing object identities.
- Do not add new furniture or remove existing items.
- Keep the target object in exactly the same pose, position, orientation, scale,
↪ material,
and appearance as in the original scene.
- Do not move, rotate, deform, lift, translate, or reposition the target object in any
↪ way.
- Only add the human in plausible contact, as if about to start the action in the next
↪ frame.

Output:

- Return only the edited scene image.

A.1.3 Best Human-Frame Selection System Prompt Π_{sel}

PLACEHOLDER: Final best-frame selection system prompt not fixed yet.

Intended role:

- Given the original scene image, the interaction description or SIG description, and
↪ multiple human-inpainted candidate frames, select the single candidate used as the
↪ visual prior for GVHMR initialization and contact reasoning.

Intended ranking criteria:

1. Action expression: the human should clearly express the requested interaction or
↪ contact-ready pre-action state.
2. Target-object preservation: the target object should remain fixed in pose,
↪ location, scale, material, and appearance relative to the original scene.
3. Plausible body pose: the inserted human should have a physically plausible
↪ full-body pose, orientation, scale, and support.
4. Contact readiness: the relevant body parts should appear close to or in natural
↪ contact with the intended target object surfaces or floor support regions.

Intended output:

- Return the index or identifier of the best candidate and a concise reason for the
↪ selection.

A.1.4 Agentic Contact-Overlay System Prompt Π_{con}

Provided Images:

Reference Image: An image showing a human interacting with the target object. Use this
↪ image only to infer which human body parts are physically contacting the target
↪ object and which object-surface regions are being touched.

Canvas Image: An image of the same scene without the human. Use this image as the base
↪ image and as the authoritative reference for the target object's final position,
↪ pose, geometry, and boundaries.

Task Description:

Generate an edited copy of the Canvas Image with solid-color segmentation masks
↪ overlaid only on the target object surface regions that are contacted by the human
↪ in the Reference Image.

Infer the contacted region relative to the target object in the Reference Image, then
↪ place the corresponding contact mask on the matching object region in the Canvas
↪ Image.

Critical Base-Image Rule:

Use the Canvas Image as the only valid base image. Preserve the Canvas Image exactly
↪ and only add the specified solid-color contact masks. Do not use the Reference
↪ Image as the base. Do not redraw, regenerate, stylize, relight, crop, pad, resize,
↪ move, or otherwise alter the Canvas Image.

Analysis Requirements:

1. Contact Inference:

Analyze the Reference Image to determine which specified human body parts are
↪ physically touching the target object.

2. Object-Region Identification:

For each contacting body part, identify the specific region of the target object
↪ being touched, such as the tabletop front edge, tabletop side edge, tabletop
↪ upper surface, tabletop underside, table leg, chair seat, chair arm, cabinet
↪ handle, ladder rung, rail, shelf, handle, edge, or other relevant object part.

3. Object-Relative Localization:

Map each inferred contact region onto the corresponding part of the target object
↪ in the Canvas Image. Use the Canvas Image object pose and boundaries as ground
↪ truth. Prioritize correct placement on the object in the Canvas Image over
↪ exact pixel alignment with the Reference Image.

4. Contact Footprint Generation:

Create a localized contact mask only on the object surface where contact occurs.
↪ The mask should represent the apparent contact footprint on the object, not the
↪ full visible human body part.

Execution Requirements:

Base Image:

Return the Canvas Image with the contact masks overlaid on top of it.

Canvas Size:

Match the Canvas Image dimensions and coordinate system.

Mask Placement:

Place each mask on the corresponding target-object region in the Canvas Image, using
↪ object-relative correspondence rather than direct coordinate transfer.

Mask Shape:

Use precise, dense, solid-color segmentation masks shaped like the localized contact
↪ footprint on the object surface. The mask should reflect the approximate size,
↪ orientation, and extent of the contact implied by the touching body part.

Object-Surface Clipping:

All mask pixels must be clipped strictly to the target object boundary in the Canvas
↪ Image. No mask may extend onto the human body, floor, wall, background, or any
↪ non-target-object region.

Scene Preservation:

Use the Canvas Image as a fixed camera reference. Preserve the camera pose, field of
↪ view, perspective, resolution, aspect ratio, room geometry, lighting, shadows,
↪ textures, and all existing object identities.

Keep the target object in exactly the same pose, position, orientation, scale,
↪ material, and appearance as in the Canvas Image. Do not move, rotate, deform,
↪ lift, translate, resize, crop, pad, or reposition the target object or any other
↪ scene element in any way.

Only add the specified solid-color contact mask overlays. Leave every non-overlay
↪ pixel visually unchanged from the Canvas Image.

Colors:

Use only the exact specified solid colors for the contacting body-part masks. Do not
↪ use transparency, antialiasing gradients, shadows, blended colors, or pastel
↪ variants.

Restrictions:

Do not add the generated human body.

Do not copy or paste the full visible hand, foot, hip, arm, leg, or other body-part
↪ silhouette.

Do not transfer masks by raw Reference Image pixel coordinates.

Do not add text labels, arrows, legends, boxes, keypoints, outlines, or any other
↪ annotations.

Show only the original Canvas Image with solid localized contact masks overlaid on the
↪ target object.

Example Guidance:

If the Reference Image shows a person gripping a ladder, infer which rungs or rails

↪ the hands and feet contact. Then place solid masks on those same ladder regions in

↪ the Canvas Image, aligned to the ladder as it appears in the Canvas Image.

A.1.5 Contact VLM Evaluation System Prompt Π_{eval}

You are evaluating colored contact-region masks for a human-object interaction task.

You are given three images:

Image 1: Reference Image.

This image shows a human physically interacting with the target object. Use it to
↪ infer which body parts are touching which object surfaces.

Image 2: Original Canvas Image.

This image shows the same scene without the human. This is the authoritative geometry
↪ and base image.

Image 3: Current Composite Image.

This is the original Canvas image with colored contact masks pasted onto it. Evaluate
↪ only the colored masks.

Target object:

{target_object}

Color mapping:

{color_mapping}

Task:

Decide whether the colored masks in the Current Composite Image correctly mark the
↪ localized contact regions implied by the Reference Image.

Focus especially on location. For each mask, check whether it is on the correct object
↪ part and correct local surface.

Evaluate these possible errors:

1. Missing contact: a body part touches the target object in the Reference Image but
↪ its mask is absent.
2. Extra contact: a mask exists for a body part that does not touch the target object.
3. Wrong color: a contact uses the wrong color for the body part.
4. Wrong object surface: the mask is on the wrong object part, such as wrong rung,
↪ wrong rail, wrong edge, wrong handle, wrong shelf, wrong tabletop region, etc.
5. Shifted mask: the mask is too far left, right, up, or down from the correct contact
↪ location.
6. Size error: the mask is too large or too small for the local contact footprint.

For every problem, write a concrete correction instruction for the next ChatGPT
↪ image-generation prompt.

Good correction examples:

- "Keep the original canvas unchanged and move only the green right-foot mask upward
↪ onto the lower ladder rung."
- "The yellow left-hand mask is missing. Add a small yellow mask on the vertical side
↪ rail where the left hand contacts the ladder."
- "The blue foot mask is on the wrong rung. Move it down to the rung directly under
↪ the visible foot contact."
- "The red right-hand mask is too large. Keep its center but make it smaller and more
↪ focused."
- "This mask is correct. Keep it unchanged."

Bad correction examples:

- "The result is bad."
- "Fix the mask."
- "Try again."

Return strictly valid JSON only:

```
```json
{
 "done": false,
 "confidence": 0.0,
 "correction_instruction": "text"
}
```
```

Set "done": true only if all required contacts are present, there are no extra masks,
↪ each mask uses the correct color, each mask is on the correct target-object
↪ surface, and each mask has acceptable location and size.